NOT MEASUREMENT SENSITIVE

MIL-PRF-38807C (USAF)
29 November 1996
SUPERSEDING
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10 April 1991

PERFORMANCE SPECIFICATION

TECHNICAL MANUALS - ILLUSTRATED PARTS BREAKDOWN

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the requirements for the preparation of Illustrated Parts Breakdown (IPB) Technical Manuals (TM). The TMs are to be prepared in single or subdivided format, or as a chapter of a maintenance/operation or overhaul manual, as specified by the acquiring activity (see 6.2). In addition to "paper" delivery, this specification provides for electronic delivery of data through use of the Document Type Definition (DTD) contained in Appendix A.

1.2 Detail. The level of detail contained in this performance specification is necessary to comply with the requirements of the Joint Computer-aided Acquisition and Logistics Support (JCALS) system.

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Det 2, HQ ESC/AV-2, 4027 Col Glenn Hwy, Suite 300, Dayton, OH 45431-1672, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A AREA TMSS

Distribution Statement A. Approved for public release; distribution is unlimited.

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-87929	-	Manuals, Technical: Operation and Maintenance Instructions in Work Package Format (for USAF Equipment)

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-1808	-	System Subsystem Sub-subsystem Numbering
MIL-STD-38784	-	Manuals, Technical: General Style and Format Requirements

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

PUBLICATIONS

FEDERAL CATALOGING HANDBOOK

H4/H8 Series	-	Commercial and Government Entity Codes

AIR FORCE TECHNICAL MANUALS

TO 00-25-195	-	Source Maintenance and Recoverability Coding of Air Force Weapons, Systems, and Equipment
TO 00-25-234	-	General Shop Practice Requirements for the Repair, Maintenance, and Test of Electrical Equipment

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402 or the Defense Logistics Service Center, ATTN: DLSC-WXA, Federal Center, Battle Creek, MI 49017-3084.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS.

3.1 Preparation. The general manner of preparation shall be in accordance with the requirements of MIL-STD-38784. IPB manuals shall be prepared in the 8.5 by 11-inch size. The contractor shall use the DTD in Appendix A for electronic delivery of this manual (see 6.2).

3.2 Manual layout. When the maintenance instructions for the equipment are in a single manual, the IPB shall also be in a separate single manual, except as specified in 3.2.1 and 3.2.2. All parts of the end item shall be listed in accordance with the requirements found elsewhere in this specification, regardless of the level of coverage (organizational, intermediate, or depot) of the associated maintenance manual(s).

3.2.1 Combined manuals. When specified (see 6.2), the IPB shall be included in the maintenance instructions manual as the last chapter or last Work Package (WP) and numbered in accordance with MIL-PRF-87929. However, if difference data sheets are to be included, the IPB shall precede them. The Chapter 1 material shall be included in the manual foreword or, when using MIL-PRF-87929 format, in the IPB WP foreword. The Chapter 3 and 4 material shall follow the Maintenance Parts List (MPL). Combined manuals shall not exceed 800 page units (including foldouts).

3.2.2 Subdivided manual. When specified (see 6.2), the IPB manual MPL chapter shall be subdivided to follow the chapter or WP structure of the maintenance manuals. Each subdivision shall be assigned a separate Technical Order number, for example, 1X-XX-4-1, 1X-XX-4-2, 1X-XX-4-3, etc. Chapters 1, 3, and 4 shall be combined in the first TM if it will not cause the manual to exceed 800 pages. The sequence shall be Foreword, Numerical Index, and Reference Designation Index. If combining chapters would cause the first TM to exceed 800 pages, Chapter 1 shall be included in the first TM and Chapters 3 and 4 in the last TM.

3.3 Arrangement. The manual shall contain as applicable:

Front Matter

Chapter 1 - Foreword

Chapter 2 - Maintenance Parts List

Chapter 3 - Numerical Index

Chapter 4 - Reference Designation Index and System Subsystem Sub-subsystem Number (SSSN) Index if used

3.4 Front matter. Front matter shall be prepared in accordance with the requirements of MIL-STD-38784 with the following exceptions:

3.4.1 Table of contents. In lieu of a separate list of illustrations, illustration titles shall be a part of the Table of Contents. Under the "Chapter 2 Maintenance Parts List" heading in the Table of Contents, the title of each illustration and its appropriate page number shall be shown. The illustrations shall be listed by title in alphabetical order. When portions of an IPB are divided into groups, systems, etc., illustration titles shall be listed below group, system, etc., titles. For combined manuals, the illustrations shall be listed in the manual's list of illustrations, if used. Otherwise, they shall be listed as previously described.

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3.4.2 Foreword. Foreword/preface/introduction requirements of MIL-STD-38784 shall be contained in Chapter 1, Foreword.

3.4.2.1 List of related publications. In addition to the requirements of MIL-STD-38784, separate IPB manuals for accessories, components, assemblies, vendor items, etc., shall be listed. The list shall include the part number and listed in alphanumeric sequence by part number. Model or type number shall be used when no part number has been assigned. If applicable, the publication number and title for a power package buildup manual with IPB shall also be listed.

3.5 Chapter 1, Foreword. Chapter 1 shall contain the following:

3.5.1 Model(s) covered. A listing of the models, types, configurations, modifications, task numbers, and series or blocks of the end item covered by the manual.

3.5.2 Serialization. The serialization table, when required by the acquiring activity, shall be used to specify "Usable On" codes (see 6.2).

3.5.3 Finding part numbers, illustration, description. When the MPL exceeds 16 pages, an illustrated explanation (see figure 1) shall be included to describe:

a. How to find a part number or description using the Table of Contents and illustration titles.

b. How to find the illustration or description if a part number or reference designation is known.

3.5.4 Listing of similar assemblies. An explanation of the method used to list parts of similar assemblies shall be included (see 3.6.9.7).

3.5.5 Parts in kits or quick change units. When repair parts are supplied in the form of kits or quick change units, an explanation shall be included, and shall include information on determining what parts are included.

3.5.6 Symbols. An explanation of the dash (-) (see 3.6.7.1.4), number sign (#) (see 3.6.9.2 and 3.6.10.3), asterisk (*) (see 3.6.9.9), and "F" (see 3.7.2.2) symbols shall be included.

3.5.7 Sheet number explanation. If multisheet illustrations are used, an explanation shall be included that the sheet number follows the index number and is separated with a slash (/) (see 3.6.7.1). The explanation shall also state that, if an indexed item is on more than one sheet of an illustration, the first and last sheet numbers on which the item appears are given.

3.5.8 Usable on codes. Definitions of the usable on codes used in the MPL of the manual shall be provided. An explanation that codes are omitted for assemblies and parts installed in all configurations of the end items covered in the manual shall be included. Codes shown shall not be limited to the initial production configuration of a part or assembly.

3.5.9 Source, Maintenance, and Recoverability (SMR) codes. The following statement shall be included: "This manual contains Joint Military Service Uniform Source, Maintenance, and Recoverability (SMR) codes. Definitions of these SMR codes are available in TO 00-25-195."

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3.5.10 Nuclear hardness. If equipment covered has nuclear survivability requirements (for example, Over Pressure and Burst, Thermal Radiation, Electromagnetic Pulse, or Transient Radiation Effects on Electronics), all Hardness Critical Items (HCI) shall be marked. Unless otherwise specified by the acquiring activity, all HCIs shall be marked with the symbol HCI immediately preceding the description of the item (see 3.6.7.4.1.). When approved by the acquiring activity, the symbol **HCI** may be used in lieu of the boxed HCI symbol (see 6.2).

3.5.10.1 Nuclear hardness symbol explanation. Chapter 1 shall include a listing and explanation of the HCI symbol and other pertinent information, as necessary, to emphasize the uniqueness of hardness features. This shall include an explanation that the HCI symbol establishes special requirements limiting changes and substitutions, and that the specific parts listed must be used to ensure hardness is not degraded. This statement shall be preceded by a CAUTION heading. All changes to, or proposed substitutions of, HCIs must be approved by the acquiring activity (see 6.2).

3.5.11 Electrostatic Discharge Sensitive (ESDS) parts. If equipment covered contains ESDS parts, they shall be marked. Unless otherwise specified by the acquiring activity, all ESDS parts shall be marked with the ESDS symbol immediately preceding the description of the item (see 3.6.7.4a, 6.2). When approved by the acquiring activity, the symbol **ESD** may be used in lieu of the ESDS symbol (see 6.2).

3.5.11.1 ESDS symbol explanation. Chapter 1 shall include a listing and explanation of the ESDS symbol: Other pertinent information shall be included as necessary to emphasize the uniqueness of ESDS parts. This shall include an explanation that the ESDS symbol requires that all ESDS parts be handled in accordance with the ESDS device handling procedures in TO 00-25-234. This explanation shall be preceded by a CAUTION heading.

3.5.12 Parts standardization. The following paragraph shall be included in the foreword chapter:

> "Parts Standardization. Authority for use of a part number different than the part number listed in this IPB is established by the Department of Defense (DoD) Interchangeability and Substitution (I&S) Program. Refer to the DO43B Master Item Identification Base for Air Force I&S information. The maintenance technician has final responsibility and authority for determining acceptability of substitute parts."

3.5.13 Manufacturers list. When required by the acquiring activity, a manufacturers list shall be provided (see 6.2). The list shall contain the Commercial and Government Entity (CAGE) code and the name and address of all manufacturers whose CAGE codes are referenced in the MPL.

3.6 Chapter 2, Maintenance parts list. The MPL shall contain:

> a. A breakdown of the assemblies and parts contained in the end item(s) or sold to the government as a peculiarly configured spare. These assemblies can be disassembled, repaired, obtained from reclamation, manufactured, reinstalled, replaced, or reassembled, in agreement with government assigned Source, Maintenance, and Recoverability (SMR) coding (see 3.6.5 for exceptions). The listings shall follow the same sequence they are discussed in the applicable maintenance manual. If the maintenance manual discusses the compressor, augmenter, and accessories, in that sequence, the MPL shall list these components and their detail parts in the same sequence (see 3.6.7 for additional information on the sequence of listing parts).

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b. Oversize and undersize parts, attaching parts, undrilled/untrimmed parts, repair parts kits, quick change units, decalcomanias, markings, etc.

c. Special support equipment, other than test equipment, which must be designed and developed in conjunction with development of the end item. Be locally manufactured for use with the end item for which no separate IPB manual has been procured. When such equipment is source coded MO, MF, or MD in the SMR column, the drawing number shall be included in the description column.

d. Identification of all HCI and ESDS components using the appropriate symbols (see 3.5.10 and 3.5.11).

3.6.1 Margin data. Margin data shall be in accordance with the requirements of MIL-STD-38784. When the MPL is to be used in conjunction with MIL-PRF-83495 type manuals, the MPL shall display the applicable SSSN, in 18 point type, in the lower outer corner of each page. The SSSN shall be placed directly above the page number.

3.6.2 Nomenclature. When SSSNs are used, the equipment and nomenclature shall be followed by the applicable higher level designation in parentheses. Chapter/section titles and system, subsystem, or sub-subsystem nomenclature shall be followed by the applicable SSSN in parentheses.

3.6.3 Front matter. This chapter shall not contain front matter unless the IPB is a subdivided manual. If subdivided, each subdivision shall have a Title Page, List of Effective Pages, and Table of Contents.

3.6.4 Parts to be listed. Each item (e.g., detail, part, assembly, etc.,) shall be listed as follows:

a. An item covered by an approved standard, and used without alteration or selection, shall be identified by the standard part number (such as MS number of Military Specification Sheets or Military Sheet Form Standards).

b. An item covered by an approved government specification containing a part identification system, used without alteration, shall be identified by that specification part identification.

c. All other items shall be identified by a part number and CAGE code (see note 1).

d. Design activities using items other than their own design, without alteration or selection, shall identify such items by the original design activity part number and CAGE code.

e. Items which are altered, selected, or source controlled, shall be identified by a part number, established by the using design activity, and CAGE code.

f. Item(s) of supply part number(s) and CAGE code that is stocked, stored, and issued by the government, as identified by the government during the initial provisioning process. When items of supply have not been identified in time for publication in the initial issue, subsequent updates shall incorporate the identified items of supply received from the government.

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g. Only the preferred or master part shall be listed. IPBs shall not be updated, changed, or revised solely to reflect a new preferred or master part number (see notes 2 and 3).

NOTE 1: A specification control drawing number (with any applicable dash numbers, if tabulated) is a part identification number used for control purposes only, and shall not be listed, but shall be part of the description.

NOTE 2: When an original assembly or part does not have continued application (no spares of the original were procured or such spares are no longer authorized for replacement), only the preferred assembly or part is listed.

NOTE 3: When an assembly or part was installed during modification, and the original part does not have continued application, only the preferred item is listed (see 3.6.7.6.1)

3.6.5 Parts not to be listed. The MPL shall omit:

a. Parts which lose their identities by being welded, sealed, or joined to other pieces as a permanent assembly.

b. Parts made of bulk stock, such as safety wire, bonding braid, upholstery cloth, friction tape, electrical wire, insulation, etc.

c. Structural items, such as stringers, stiffeners, skin, etc. (Applicable only to IPB manuals for aircraft and missiles).

d. Commonly used hardware, such as standard rivets, screws, bolts, nuts, cotter pins, washers, etc., when not used as attaching parts for frequently detached assemblies, subassemblies, or detailed parts.

e. Support equipment, such as tools and test equipment, other than that defined in 3.6c.

3.6.6 Illustrations. Illustrations shall be provided to identify listed parts. Illustrations shall be selected, prepared, and located in accordance with the requirements of MIL-STD-38784, and as specified herein. When necessary, an illustration shall show the relation of a component to the end item. When identical parts appear in the same relative location, only one need be illustrated. Exploded views shall be used to portray disassembly sequence clearly or to identify parts rapidly. Every effort shall be made to limit the number of parts shown on an illustration, so that no more than one page is required. If multisheet illustrations are required, they shall be in accordance with MIL-STD-38784. Foldout pages shall not be used.

3.6.6.1 Main groups/main assemblies. When the MPL is divided into main groups or major assemblies of the end item, the first illustration shall portray this information with a key to the main groups or major assemblies (see figures 2 and 3).

3.6.6.2 Functional systems. A functional system (air conditioning, communications, flight controls, fuel, hydraulic power, landing gear, etc.) shall be illustrated in a manner that permits the individual assemblies of the system to be readily located in the MPL by the use of keys (see figure 4).

3.6.6.3 Exploded view. Figure 5 provides a typical exploded view for equipment.

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3.6.6.4 Similar assemblies and parts. When assemblies and parts have the same appearance and functional purpose, they need to be illustrated only once, provided the criteria of 3.6.9.7 are met.

3.6.6.5 Differing components, assemblies, and parts. Components, assemblies, and parts which differ from an earlier or later configuration, yet have a similar appearance, need to be illustrated only once, providing the changed and added parts are clearly identified. When space does not permit this arrangement, or major differences exist, separate illustrations shall be provided.

3.6.6.6 Subassemblies. When space permits, all the detail parts of each subassembly contained in any one assembly shall be shown on one illustration. When space limitations prohibit this requirement, a subassembly shall be illustrated completely assembled on the assembly illustration, and a separate exploded view of the subassembly prepared. In both descriptions of the subassembly (where listed assembled and where listed with detail parts), there shall be a cross-reference to the other illustration.

3.6.6.7 Attaching parts. Illustrations of attaching parts shall only be exploded when disassembly/assembly procedures merit explosion because of complexity. When the attaching parts at a specific location are not visible in an illustration (a top view may show the bolts and washers above, but not the washers and nuts beneath), an index number shall be assigned to each item, and all index numbers shall be shown with one leader line terminating on the visible part(s).

3.6.6.8 Indicating relationship. When unlisted parts are illustrated to show relationship, the unlisted parts shall be toned down, or shown in phantom, to give emphasis to listed parts.

3.6.6.9 Index numbers. Each illustration shall have index numbers beginning with 1. Any suitable method which will effectively identify a part may be employed, such as use of leader lines, index number directly on a part, etc. Index numbers must agree with those shown in the listings (see 3.6.7.1 and 3.6.7.1.1). When disassembly/assembly procedures do not merit explosion because of simplicity (for example, the insulating pad or socket for a microcircuit), an index number shall be assigned to each item, and all index numbers shall be shown with one leader line terminating on the visible part(s) (see figure 6). When specified by the acquiring activity, reference designators shall follow or be placed immediately below the applicable index number, in parentheses, on the illustration (see 6.2).

3.6.6.10 Figure titles and numbering. The requirements of MIL-STD-38784 are applicable except the chapter number shall not precede the figure number, unless the IPB is a chapter of a maintenance manual.

3.6.6.11 Added and deleted figures and index numbers. The requirements of MIL-STD-38784 apply.

3.6.6.12 Circuit card assemblies (CCA). CCAs and their components shall be illustrated and broken down in accordance with SMR code assignments (maintenance concept), as established by the acquiring activity. If the CCA is of such a nature that it cannot be adequately illustrated by a single indexed view, it shall be indexed by sections as separate views. In such cases, a view showing the complete CCA, and identifying each section, shall be shown. Each section shall then be indexed (see figure 6). Exploded views shall be used when necessary to clearly portray all indexed components of the CCA.

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3.6.7 Format (see figures 2, 3, 4, 6, and 7). The MPL shall be arranged in the format indicated below, and shall not be right justified. The parts listing shall be considered part of the associated figure, and shall not be separately labeled (e.g., an illustration of the filter assembly and its associated parts list would form "Figure X. Filter Assembly."). The width of the columns (description column excepted) shall be adjusted to accommodate the information to be inserted in each; the remaining space shall form the description column. When specified by the acquiring activity, if the end item consists of main groups and/or systems, the title of each main group or system shall be shown beneath the publication number (upper outer corner) on each applicable page (see 6.2). Parts shall be listed to show parts relationship and in the sequence of disassembly, except where that sequence cannot be maintained.

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1234567	UNITS PER ASSY	USEABLE ON CODE	SMR CODE

3.6.7.1 Figure & index/sheet no. column. This column shall list applicable figure, index, and sheet numbers (see 3.6.7.1.2). The figure and index numbers shall be separated by two spaces. The index and sheet numbers shall be separated by a slash (/). Figure 12, index 1, on sheet two of the illustration would be listed as "12 1/2" The figure number shall precede the first index number in each listing and the first index number on each page of a multi page listing, but shall not be repeated before each subsequent index number. Contractors shall ensure that the index numbers in the parts list are in sequence. When the same part appears more than once on an illustration, and is part of different assemblies or subassemblies, or it is otherwise necessary to list it more than once, each listing shall be assigned a different index number. An index number shall not be assigned to an assembly when the detail parts are indexed unless the assembly is shown completely assembled on the same illustration, or a bracket or circle shows which parts comprise the assembly. When a change to an end item requires addition of parts to a listing and illustration, the requirements of MIL-STD-38784 apply for suffixed index numbers.

3.6.7.1.1 Index numbers for CCAs. For CCAs (because of the many like items); when the same part appears more than once in an illustration, the same index number shall be assigned to all identical parts.

3.6.7.1.2 Index numbers for attaching parts. When a group of parts (bolt, washer, nut) is used at a specific location for attachment purposes, each part shall be assigned a separate index number (see 3.6.6.7).

3.6.7.1.3 Sheet numbers. Sheet numbers shall be used only with multisheet illustrations. When an indexed item appears on more than one sheet of the illustration, the first and last sheet number on which the item appears shall be listed.

3.6.7.1.4 Parts which are not illustrated. Parts which are indexed in the listing, but are not shown on the illustration (e.g., attaching parts or a microcircuit socket which are not exploded (see 3.6.6.7 and 3.6.6.9)) shall have a dash (-) preceding the index number (e.g., -47). Chapter 1 shall include an explanation that the symbol denotes a part which is not illustrated.

3.6.7.2 Part number column. This column shall contain part numbers, including dash numbers, assigned to each listed part in accordance with current industry engineering practices. Only the preferred or master part shall be listed (see 3.6.4). As applicable, part numbers for prime

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contractor, subcontractor, vendor, commercial parts, and standard (government, contractor and industry) parts shall all be listed. When the part number and drawing number differ (excluding dash numbers), refer to 3.6.7.4d. When a component is identified only by a model or type number, such a number shall be used in lieu of a part number.

3.6.7.3 Commercial and Government Entity (CAGE) column. The appropriate CAGE code, published in current issues of H4/H8 Cataloging Handbooks, shall be listed in this column directly opposite each part, model, or type number listed in the part number column. The CAGE code shall identify the design activity or government agency whose number appears in the part number column.

3.6.7.3.1 CAGE code not listed in H4/H8. When a CAGE code for the appropriate design activity or government agency is not published in the current issues of the H4/H8 Cataloging Handbooks, the word "none" shall be inserted in the CAGE column directly opposite the part, model, or type number listed in the part number column (see 3.6.7.4h).

3.6.7.4 Description column. This column shall contain the description of each part listed. The nomenclature appearing in the title block of the drawing, to which the detail part was manufactured, shall be used. Incomplete nomenclature (e.g., noun only) of detail parts, appearing in the body of an assembly or installation drawing, shall not be used (government standard parts excepted). When necessary, the size, dimensions, material, and tolerances for a part shall be indicated to make the description complete. The description shall be arranged to show the following:

a. The identifying noun shall be the first word. When the part is identified as a Hardness Critical Item (HCI) or Electrostatic Discharge Sensitive (ESDS), the HCI, and ESDS markings, as appropriate, shall precede the first word (see 3.5.10, 3.5.11 and the example following 3.6.7.4.1). When the listing is an assembly or installation, the word "assembly" or "installation" shall follow the noun. A comma shall separate these words from the balance of the description (see 6.5).

b. The item name (including the word "assembly" or "installation") shall be in upper-case. The first letter of the first word following the item name shall be in upper-case; remaining words shall be in lower-case. The first letter of proper nouns, regardless of location, shall always be in upper-case.

c. When units of measurement are the same, they shall not be repeated with each dimension; example: 1/8 by 3/4-inches (not 1/8-inch by 3/4-inch). To avoid confusion, a dash shall be used between a whole number and a fraction; example: 1-1/16, 2-3/4, 4-9/64. Dashes may be omitted if fraction characters (e.g., ½, ¼) are used. When a decimal with a value of less than 1.0 is given, a zero shall precede the decimal point; example: 0.002-inch.

d. The appropriate numbers for the type, model, specification, specification control drawing, and the source control drawing shall follow the nomenclature. When the drawing number for a part is different than the part number (excluding dash number differences), the drawing number shall follow the description (see 3.6.7.2). When a part is authorized for local manufacture, and a specific drawing is required for such accomplishment, the drawing number shall follow the description.

e. The requirements of MIL-STD-38784 apply to the use of abbreviations.

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f. Leader points (a series of periods) shall be used to join the description and the next column to the right. When the description requires more than one line, leader points shall be used on the first line only. Leader point spacing shall be at least double-spaced.

g. Reference to another illustration for information on the detail parts of an assembly and reference to another illustration for information on the next higher assembly shall be made.

h. When the word "none" is listed in the CAGE code column in accordance with 3.6.7.3.1, the complete name and address of the design activity or government agency, whose part, model, or type number appears in the part number column, shall be shown in parentheses at the end of the description. When several CAGE codes are not available, or the design activity's name and address must be repeated several times, numerically identified footnotes may be used in the CAGE column and identified at the end of the MPL and in Chapter 1.

3.6.7.4.1 Indentation. The description for parts listed shall be indented to indicate relationship. The indentation shall be indicated by leader points, each point shall signify each indentation. When the description exceeds one line, the second (and succeeding lines) shall be indented two additional indentations from the first line.

Description

1 2 3 4 5 6 7

END ITEM, COMPONENT, MAJOR ASSEMBLY
. HCI DETAIL PARTS FOR END ITEM, COMPONENT, MAJOR ASSEMBLY
. ASSEMBLY
. Attaching parts for assembly (AP)
. . Detail parts for assembly
. . Subassemblies
. . Attaching parts for subassemblies (AP)
. . . Detail parts for subassemblies
. . . Sub-subassemblies
. . . Attaching parts for sub-subassemblies (AP)
. . . . Detail parts for sub-subassemblies

3.6.7.5 Units per assembly column. The column shall show the following:

a. Quantity required for each detail part in an assembly.

b. Quantity required for each assembly in the next higher assembly.

c. Quantity required for each attaching part to attach one unit or one assembly.

d. The abbreviation "AR" for oversize and undersize parts and "as required" for parts when quantities are indefinite.

e. The abbreviation "REF" where items are listed for reference purposes. Such items shall show a quantity where the item is listed assembled, or listed in detail, and show "REF" in other listings.

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f. The quantity shall be listed opposite the first line of the description when the description requires more than one line.

3.6.7.6 Usable on code column. This column shall contain codes to indicate the configurations (type, model, series, blocks, etc.) of the end items to which listed assemblies and parts apply. When a part applies to all configurations, no code is required in this column; likewise, when the manual covers only one configuration, no coding system is required. The coding system shall consist of letters of the alphabet beginning with single letters, A through Z, and when necessary, continue with double letters, AA through AZ, BA, through BZ, etc.

3.6.7.6.1 Usable on code. The meaning of Usable on Codes, if used, shall appear at the end of each MPL (for each figure) in the description column (see figure 7). Example:

CODE	USABLE ON
A	Aircraft Number/Serial Number through Aircraft Number/Serial Number
B	Aircraft Number/Serial Number through Aircraft Number/Serial Number before TCTO XX-XX-XXX
C	Aircraft Number/Serial Number through Aircraft Number/Serial Number after TCTO XX-XX-XXX

The MPL shall not list Time Compliance Technical Orders (TCTOs) that have been completed.

3.6.7.7 SMR code column. This column shall contain the five digit Joint Military Services Uniform SMR codes assigned by the government and furnished by the acquiring activity. When codes are not available for publication in the initial issue, scheduled changes or revisions, as applicable, shall incorporate SMR codes approved by the government during the initial provisioning process.

3.6.8 Layout. Whenever the size of an illustration and the length of its corresponding parts list warrant, the MPL shall be laid out so that a full-page illustration is on the left-hand page and the applicable parts list is on the right-hand facing page. Whenever a smaller illustration and parts list length permits, the illustration and parts list shall be placed on the same page; the illustration may be placed above or below the parts list.

3.6.9 Subjects requiring special attention.

3.6.9.1 Standard items and standard parts. Government standard items and parts shall be identified by the assigned government standard part number and CAGE code. Part numbers, design activity codes, and complete descriptions are required for industry and contractor standard items and parts. When an item is part of an assembly and covered in the intermediate level IPB, it must be shown in the aircraft IPB if it is SMR coded for organizational level maintenance.

3.6.9.2 Government Furnished Equipment (GFE) and Contractor Furnished Equipment (CFE) covered by separate manuals. The item and attaching parts shall be listed for both GFE and CFE. Other detail parts shall not be listed. Following the part number of the GFE/CFE item, a number sign (#) shall be inserted flush right to indicate that detail parts are listed in a separate manual. Chapter 1 shall include an explanation that the number sign signifies that detail parts are listed in a separate manual and the publication number is in the list of applicable technical manuals.

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3.6.9.3 Vendor Parts (including commercial hardware). When vendor items are used exactly as manufactured, and are not selected or source controlled, vendor part numbers shall be listed. When a separate IPB is available for a vendor item, the requirements of 3.6.9.2 are applicable.

3.6.9.4 Altered, selected, or source controlled items. When a government standard, industry standard, vendor, or commercial item is altered, selected, or source controlled, the part number of the design activity responsible for the alteration, selection, or source control shall be listed in the part number column.

3.6.9.5 Oversize and undersize parts. The description of oversize and undersize parts shall include dimensions.

3.6.9.6 Matched parts. When parts are manufactured as a matching set (lapped assembly, electronic, etc.) to meet certain requirements, the description of the part shall be coded or annotated that the parts must be requisitioned as a set.

3.6.9.7 Similar assemblies. When similar assemblies contain 51 percent or more identical parts, the assemblies shall be combined and listed as follows; otherwise the assemblies shall be listed separately.

a. All the assemblies (figure and index numbers, part numbers, description, quantities, codes) shall be listed first, followed by the detail parts.

b. A part, common to all assemblies in the same quantity, shall be listed once and the quantity per assembly designated.

c. A part, common to all assemblies in differing quantities, shall be listed once for each quantity, and be identified to which assembly each listing pertains, by use of a usable on code.

d. Peculiar parts shall be listed once, and be identified to which assembly each pertains by use of a usable on code.

3.6.9.8 Attaching parts. Attaching parts shall be identified by the abbreviation (AP) following the description of the part. Example:

Plate
Screw (AP)
Washer (AP)
Connector

3.6.9.9 Marking. Decalcomanias, metalcals, vinyl film markings, etc., shall be considered as parts. The identifying drawing number for each marking shall appear in the Part Number column. The part number for each marking shall be followed (flush right) by an asterisk (*) symbol. Chapter 1 shall explain that such a symbol means, "requisition this marking in accordance with the requirements of AFI 37-162." The nomenclature or title of each marking shall appear in the description column. The location of each marking shall be illustrated, but a separate readable illustration is not necessary.

3.6.10 Parts in kits, change units, power packages.

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3.6.10.1 Repair parts kits. When replaceable parts of an assembly are available in the form of kits, the words "(Repair kit available)" shall follow the description of the applicable assembly. The detail parts of the assembly that are contained in the kit shall be assigned kit source codes and these codes shall appear in this column. The kit listing shall follow the last detail part of the applicable assembly and be listed at the same indentation as the assembly. Illustrations of kits are not required.

3.6.10.2 Quick Change Units (QCU). When an item is supported by a QCU, the words "(QCU available)" shall follow the description of the applicable item. The detail parts of the item that are contained in the QCU shall be identified with the words "QCU detail" following the description of each detail part. The QCU listing shall follow the last detail part of the applicable item and be listed at the same indentation as the item. Following the listing of the QCU, a quick change kit, if applicable, shall be listed. Illustrations of QCUs are not required.

3.6.10.3 Power package. When the acquiring activity requires a separate power package buildup manual (including IPB) to be prepared, the end item IPB shall contain an illustration and listing of the assembled power package, and illustrations and listings of attaching parts for the power package (see 6.2). Detail parts of the power package shall not be listed. Following the part number of the power package, a number sign (#) shall be inserted flush right to indicate that detail parts are listed in a separate manual. Chapter 1 shall include an explanation that the number sign signifies that detail parts are listed in a separate manual and the publication number is in the list of applicable technical publications (LOAPS).

3.7 Chapter 3, numerical index. This chapter shall contain the part numbers of all parts listed in the MPL, including model and type numbers for components not assigned part numbers. A numerical index is not required when the MPL contains less that 200 different part numbers.

3.7.1 Front matter.

a. Single manuals. This chapter shall not contain front matter.

b. Subdivided manuals. This chapter shall have a Title Page and List of Effective Pages.

c. Subdivided manuals with Chapters 1, 2, and 3 combined. This chapter shall not contain front matter.

3.7.2 Format. The numerical index shall be arranged in the following format. Unless otherwise specified by the acquiring activity, a space shall be skipped beneath each tenth listing (see 6.2). The width of the columns shall be adjusted to accommodate the information to be inserted in each. A triple division of the format shall be used when this will not cause a cluttered or difficult to read index, in which case a double division shall be used.

PART NUMBER	FIGURE & INDEX	PART NUMBER	FIGURE & INDEX	PART NUMBER	FIGURE & INDEX

3.7.2.1 Part number columns. Part number arrangement shall begin at the extreme left and continue from left to right, one position at a time. For the first character of the part number, the letters A through N and P through Z take precedence over the numerals, 0 through 9. (Alphabetic

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Os are considered numeric zeros). For the second and succeeding characters of a part number, precedence is as follows: (1) diagonal, (2) period, (3) dash, (4) letters A through N and P through Z, (5) numerals 0 through 9. The following is a sample arrangement:

AN931-4-13	B2	16.W2
A2460	S/1	16W060
A317	1140	32P010-1
A32	121873	32P0101
B12	128	39A46

3.7.2.2 Figure and index number column. This column shall contain the figure and index number(s) for all parts listed, as applicable. When an assembly or part has not been assigned an index number, the figure and index number of the preceding part in the MPL shall be used with the letter "F" before the figure number, such as F7-6. Chapter 1 shall include an explanation that the letter "F" means "follows". For subdivided manuals, the applicable manual number shall be added to the figure and index number column.

3.8 Chapter 4, reference designator index. This index is required only when reference designators have been established for any parts listed in the MPL. When the MPL contains less than 200 different part numbers, a reference designator index is not required, provided the applicable reference designators have been inserted on the illustrations. When specified by the acquiring activity, the SSSN shall be used in conjunction with, or in lieu of, the reference designator (see 6.2). The SSSN is assigned in accordance with MIL-STD-1808.

3.8.1 Front matter.

a. Single manuals. This chapter shall not contain front matter.

b. Subdivided manuals. This chapter shall have a Title Page and List of Effective Pages.

c. Subdivided manuals with Chapters 1, 2, and 3 combined. This chapter shall contain no front matter.

3.8.2 Format. The Reference Designator Index shall be arranged as follow:

a. A space shall be skipped beneath each tenth listing.

b. The width of the columns shall be adjusted to accommodate the information to be inserted in each.

c. Format shall be three double-columns.

Reference Designator	Figure & Index No.	Reference Designator	Figure & Index No.	Reference Designator	Figure & Index No.

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3.8.2.1 Reference designator column. This column shall contain all applicable reference designators shown on schematic and wiring diagrams, and those cited in applicable operation and maintenance manuals. They shall be arranged in alphanumeric sequence. When the block or unit system of designations is employed, identical items having reference designators which are consecutive, and are at the same location, may be grouped. When use of SSSN has been specified (see 3.8), this column shall be headed as "SSSN" with the SSSNs following in numeric order.

3.8.2.2 Figure and index number column. This column shall contain the figure and index number(s) applicable to all reference designations listed. When an assembly or part has a reference designator, but not an index number, the figure and index number of the preceding part in the MPL shall be used with the letter "F" before the figure number, such as F7-6. Chapter 1 shall include an explanation that the letter "F" means "follows." For subdivided manuals, the applicable manual number shall be added to the figure and index number column. For work package type manuals, the work package number shall be included with the figure and index number.

3.9 Changes and revisions. Changes and revisions shall be prepared in accordance with MIL-STD-38784.

4. VERIFICATION.

4.1 Verification. Unless otherwise specified in the contract or purchase order:

a. Validity of the accuracy and scope of the technical content, and user interface functionality shall be the responsibility of the contractor (see 6.4.1).

b. The contractor shall provide suitable facilities to perform the validation functions specified herein.

c. The contractor's existing quality assurance (QA) procedures shall be used.

d. The government reserves the right to review any of the verifications.

4.1.1 Minimum verification requirements. As a minimum, verification shall ensure the following:

a. Suitability of the IPB for the intended maintenance environment.

b. Usability by the intended users.

c. Compatibility with other government systems.

4.1.2 Compliance. All IPB manuals shall meet all requirements of Sections 3 and 5 of this specification and the appropriate DTD appendix, as required by the acquiring activity (see 6.2). The requirements set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Use of sampling inspections shall be at the discretion of the contractor, and in accordance with commercially acceptable quality assurance procedures. However, use of sampling in QA procedures does not

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authorize submission of known defective material, either indicated or actual, nor does it commit the government to accept defective material.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Technical manuals prepared in accordance with the requirements of this specification are intended for use in the identification and requisitioning of parts, and for illustrating assembly and disassembly relationships.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this document.

b. If TMs are to be prepared in single or subdivided format, or as a chapter of a maintenance/operation or overhaul manual (see 1.1).

c. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

d. Issue of the H4/H8 Handbook to be cited in the solicitation (see 2.2.2).

e. If electronic delivery of this manual is required (see 3.1).

f. If manual is to be developed in work package format (see 3.2.1).

g. Manual layout for each IPB is to be determined jointly by the contractor and the acquiring activity based on system requirements; combined manuals are to be used to the maximum extent possible (see 3.2, 3.2.1, 3.2.2).

h. If serialization tables are to be provided whenever two or more configurations of the equipment covered by the IPB exist, and not all subcomponents are interchangeable between configurations (see 3.5.2).

i. If all HCIs are to be marked with a symbol other than that specified in this specification (see 3.5.10).

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j. If changes or substitutions of HCI are required (see 3.5.10.1).

k. If all ESDS parts are to be marked with the ESDS symbol, and if the symbol **ESD** may be used in lieu of the ESDS symbol (see 3.5.11).

l. If a manufacturers list is to be provided (see 3.5.13).

m. If reference designations are to follow, or be placed immediately below the applicable index number on the illustrations (see 3.6.6.9).

n. If MPL pages are to show group or system titles beneath the publication number (see 3.6.7).

o. If a separate power package buildup manual is being procured (see 3.6.10.3).

p. If spacing between listings is to be other than that specified in this specification (see 3.7.2).

q. If SSSN is to be used in Chapter 4 and, if used, whether it is to be in conjunction with, or in lieu of, the reference designator (see 3.8).

r. If performance of inspections is to be other than as specified herein (see 4.1).

s. Packaging requirements (see 5.1).

6.3 Acronyms. The acronyms used in this document are defined as follows:

AMSDL	-	Acquisition Management Systems and Data Requirements Control List
CAGE	-	Commercial and Government Entity
CCA	-	Circuit Card Assembly
CFE	-	Contractor Furnished Equipment
DoD	-	Department of Defense
DoDISS	-	Department of Defense Index of Specifications and Standards
DTD	-	Document Type Definition
ESDS	-	Electrostatic Discharge Sensitive
GFE	-	Government Furnished Equipment
HCI	-	Hardness Critical Item
IPB	-	Illustrated Parts Breakdown
I & S	-	Interchangeability and Substitution
JCALS	-	Joint Computer-aided Acquisition and Logistics Support
MPL	-	Maintenance Parts List
QA	-	Quality Assurance
QCU	-	Quick Change Units
SMR	-	Source, Maintenance, and Recoverability
SSSN	-	System Subsystem Sub-subsystem Number
TCTO	-	Time Compliance Technical Order
TM	-	Technical Manual
WP	-	Work Package

6.4 Definitions. For the purposes of this document, the following definitions apply.

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6.4.1 Verification. Verification (section 4), in the context of this specification equates to the contractor's quality assurance program for validating the content of the manuals and checklists. Suggested validation methods include:

a. Actual performance. Using production configured equipment, hands-on performance of the procedure using the technical instructions as written.

b. Simulation. Using production configured equipment and the manual procedures and checklists, simulate the actions required by the task steps.

c. Table top analysis. Primarily for nonprocedural data, compare the technical content to source data to ensure the technical accuracy and depth of coverage.

6.5 Description column. The noun name is adequate description for government standard parts (see 3.6.7.4a).

6.6 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.7 Subject term (key word) listing.

Commercial and Government Entity code
Maintenance Parts List
Reference Designator
Source, Maintenance and Recoverability code
Units Per Assembly
Usable On code

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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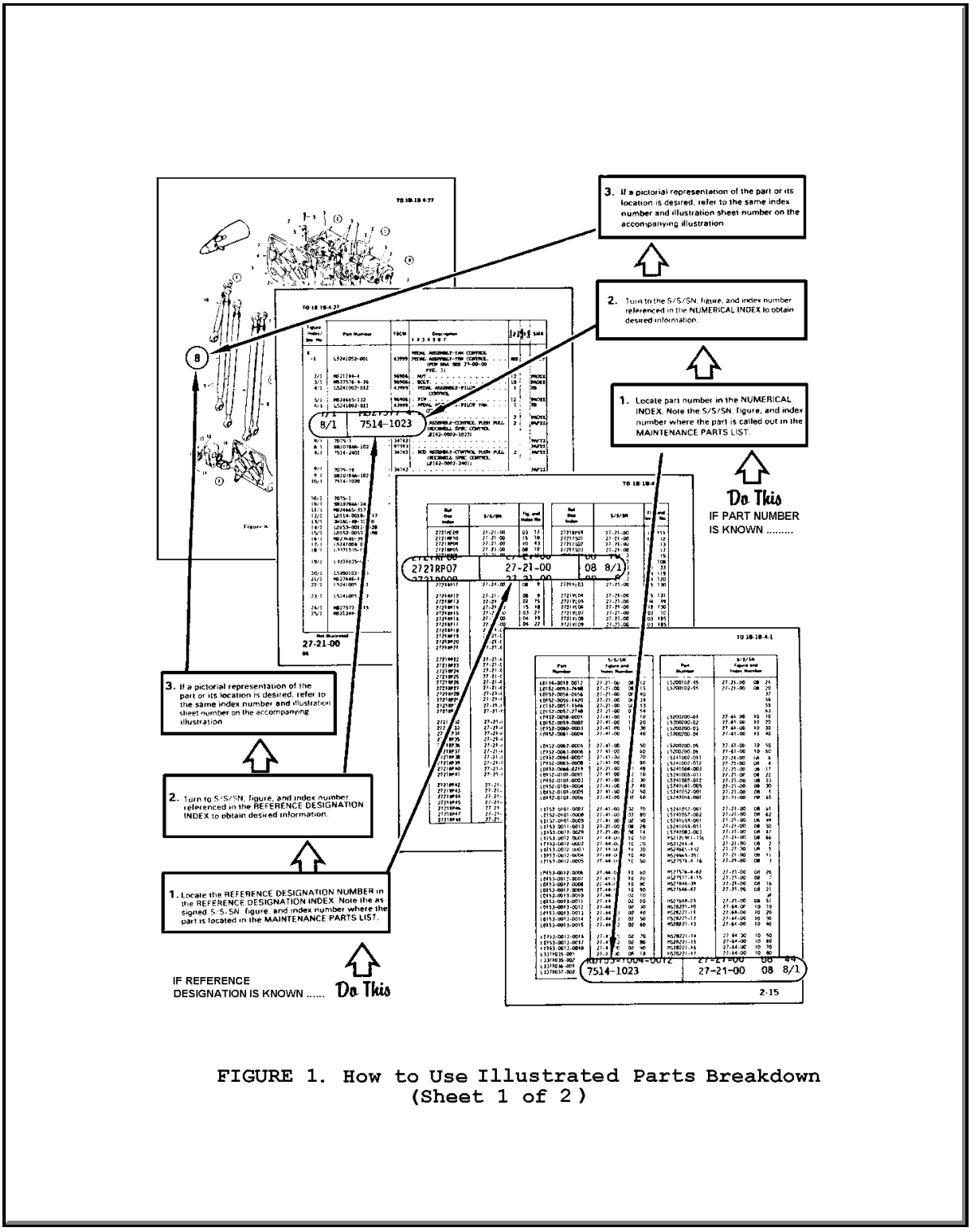


FIGURE 1. How to Use Illustrated Parts Breakdown
(Sheet 1 of 2)

FIGURE 1. Example how to use the illustrated parts breakdown.

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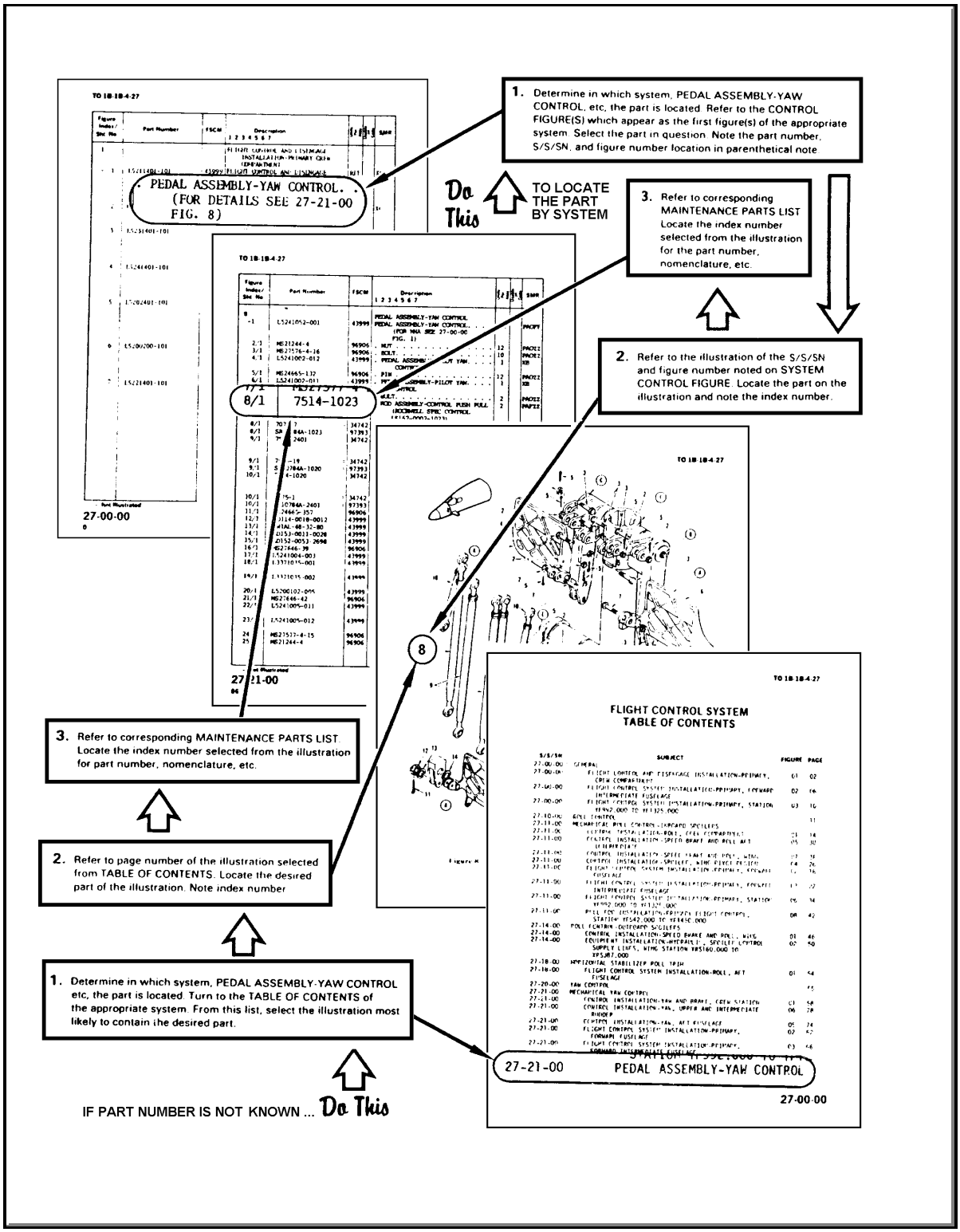


FIGURE 1. Example how to use the illustrated parts breakdown - Continued.

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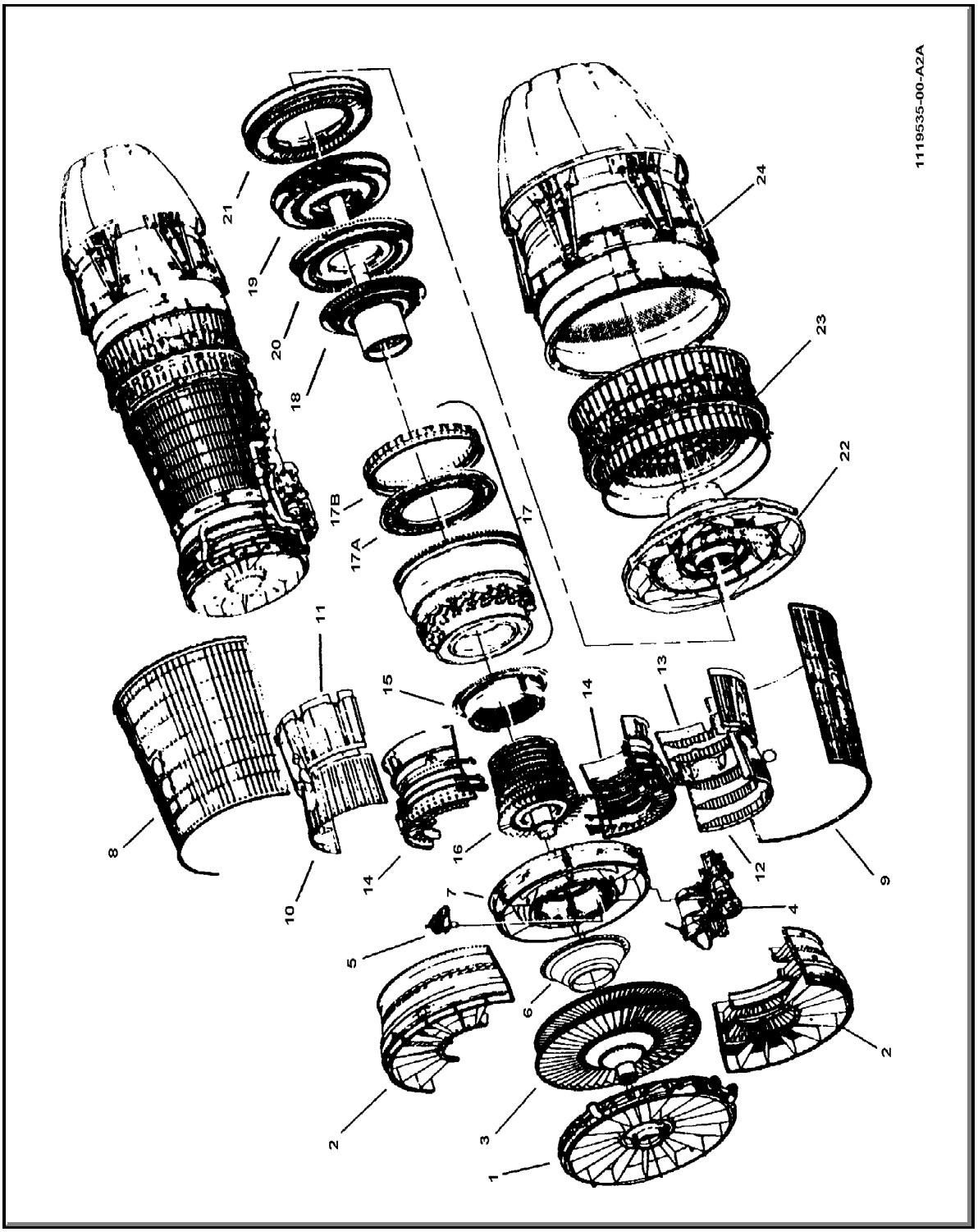


FIGURE 2. Typical main groups listing.

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FIGURE & INDEX / SHEET NO.	PART NUMBER	CAGE	1 2 3 4 5 6 7 DESCRIPTION	UNITS PER ASSY	USEABLE ON CODE	SMR CODE
			LOCATOR FOR MAJOR ENGINE ASSEMBLIES			
1 -	9550M10G01		ENGINE ASSY	1		
- 1	9550M13G03		. FRAME ASSY, FRONT/SEE FIGURE 2 THRU 5/	1		AFF
- 2	9550M11G01		. STATOR ASSY, FAN/REPLACED BY ..... 9550M11G02 PER TO 2J-F101-517 /SEE FIGURE 6 THRU 11	1		AFF
	9550M11G02		. STATOR ASSY, FAN/REPLACED BY ..... 9550M11G02 PER TO 2J-F101-517 /SEE FIGURE 6 THRU 11/	1		AFF
	9550M11G02		. STATOR ASSY, FAN/REPLACED BY ..... 9550M11G03 PER B035/REWORK AND REIDENTIFY TO 9550M11G06 PER ECP B078/SEE FIGURE 6 THRU 11/	1		AFF
	9550M11G03		. STATOR ASSY, FAN/REPLACED BY ..... 9550M11G05 PER B050/REWORK AND REIDENTIFY TO 9550M11G05 PER ECP B050/SEE FIGURE 6 THRU 11/	1		AFF
	9550M11G05		. STATOR ASSY, FAN/REPLACED BY ..... 9550M11G06 PER B078 /REWORK AND REIDENTIFY FROM 9550M11G03 PER ECP B050/SEE FIGURE 6 THRU 11/	1		AFF
	9550M11G06		. STATOR ASSY, FAN/REWORKED AND ..... REIDENTIFIED FROM 9550M11G02 PER B078/SEE FIGURE 6 THRU 11/	1		AFF
R - 3	9550M28G07		. ROTOR ASSY, FAN/REPLACED BY ..... 9550,28G11 PER ECP B140/REWORK AND REIDENTIFY TO 9550M8G11 PER ECP B140/SEE FIGURE 12/SEE FIGURE 14 FOR BREAKDOWN/	1		PAFDD
R	9550M28G08		. ROTOR ASSY, FAN/REPLACED BY ..... 9550,28G12 PER ECP B140/REWORK AND REIDENTIFY TO 9550M8G12 PER ECP B140/SEE FIGURE 12/SEE FIGURE 14 FOR BREAKDOWN/	1		PAFDD
R	9550M28G09		. ROTOR ASSY, FAN/REPLACED BY ..... 9550,28G13 PER ECP B140/REWORK AND REIDENTIFY TO 9550M8G13 PER ECP B140/SEE FIGURE 12/SEE FIGURE 14 FOR BREAKDOWN/	1		PAFDD
R	9550M28G10		. ROTOR ASSY, FAN/REPLACED BY ..... 9550,28G14 PER ECP B140/REWORK AND REIDENTIFY TO 9550M8G14 PER ECP B140/SEE FIGURE 12/SEE FIGURE 14 FOR BREAKDOWN/	1		PAFDD
R	9550M28G11		. ROTOR ASSY, FAN/REWORKED AND ..... REIDENTIFIED FROM 9550M28G07 PER ECP B140/SEE FIGURE 12/SEE FIGURE 14 FOR BREAKDOWN/	1		PAFDD
R	9550M28G12		. ROTOR ASSY, FAN/REWORKED AND ..... REIDENTIFIED FROM 9550M28G08 PER ECP B140/SEE FIGURE 12/SEE FIGURE 14 FOR BREAKDOWN/	1		PAFDD

FIGURE 2. Typical main groups listing - Continued.

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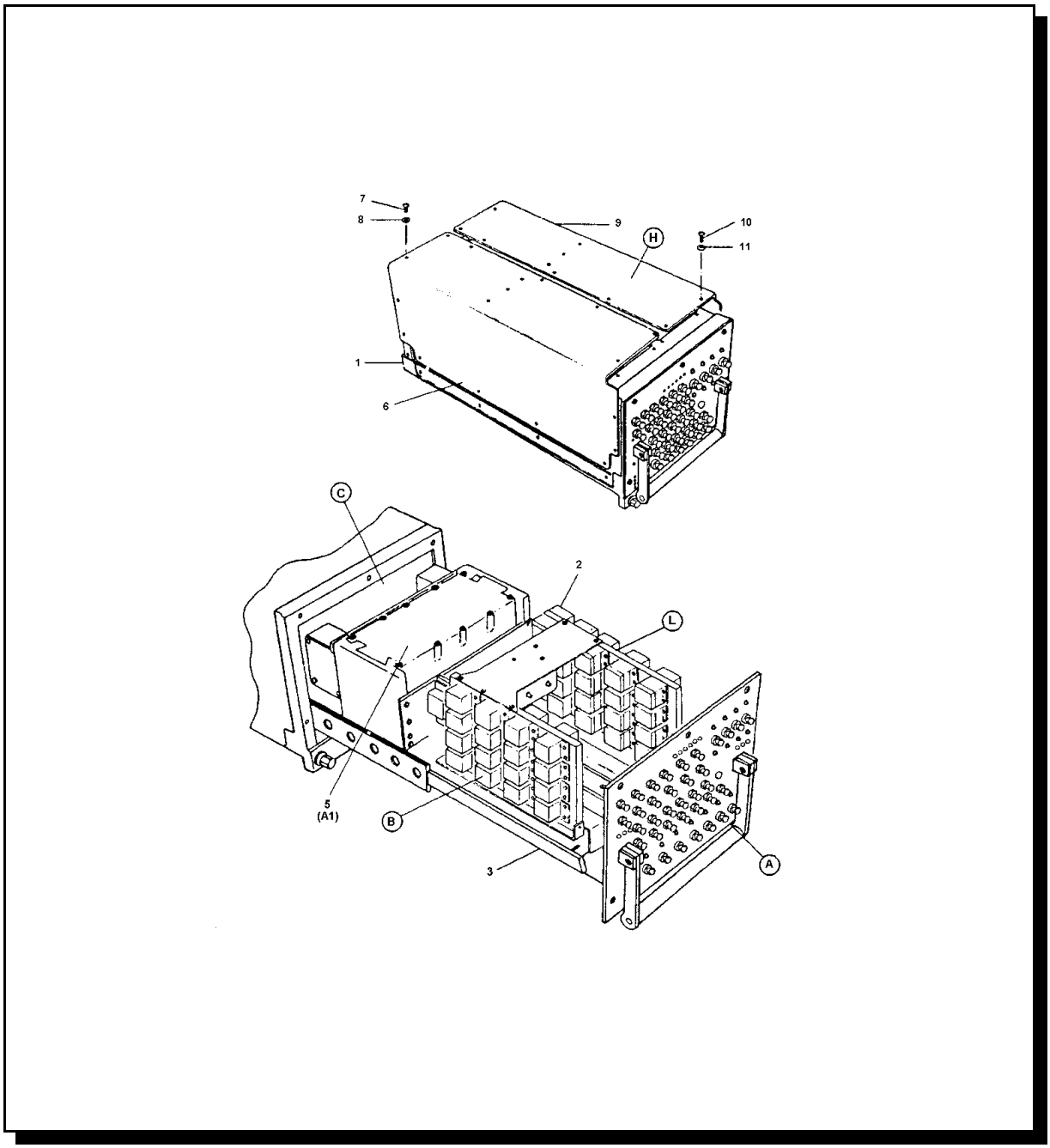


FIGURE 3. Typical main assemblies listing.

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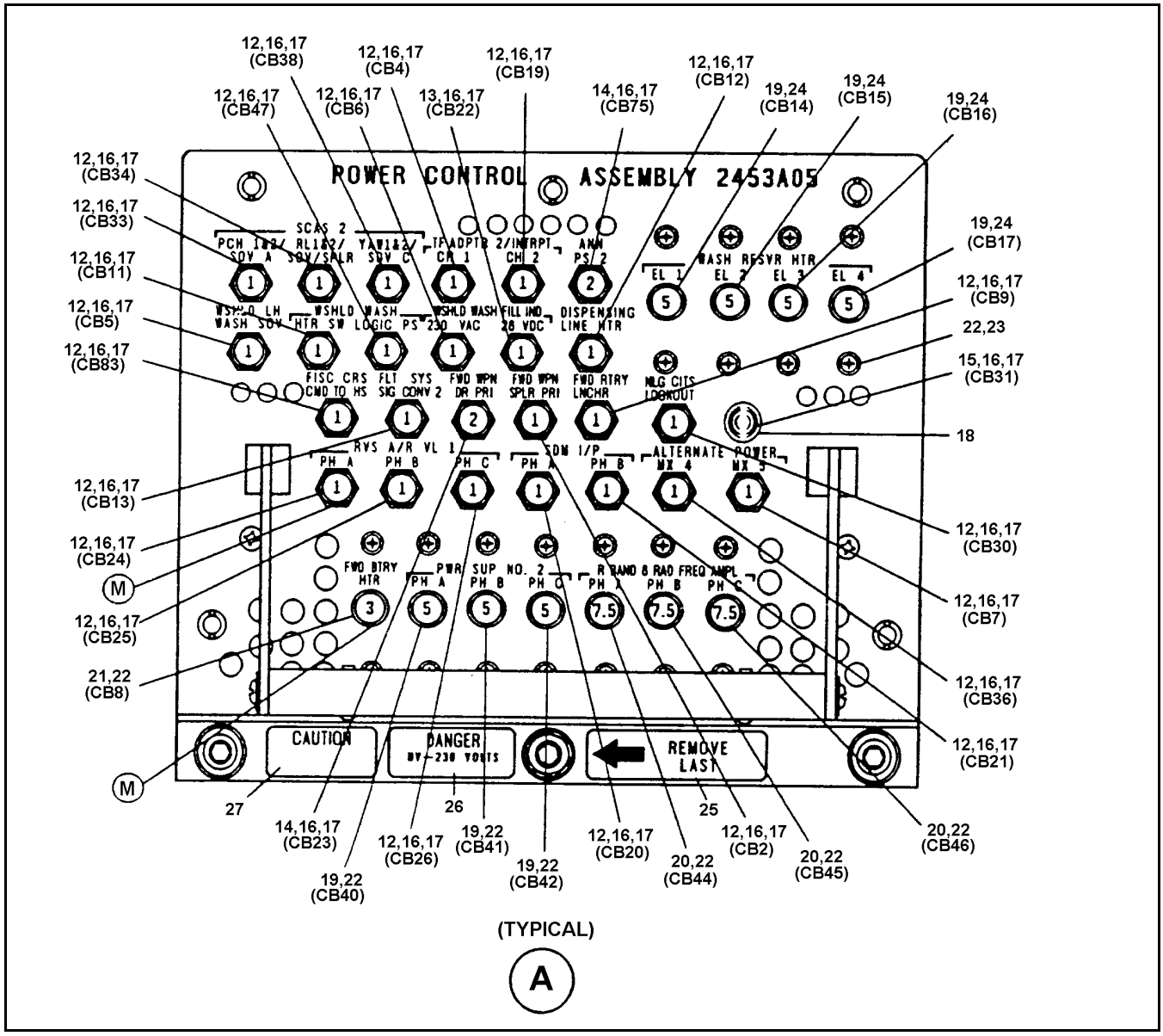


FIGURE 3. Typical main assemblies listing - Continued.

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FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USEABLE ON CODE	SMR CODE
			POWER CONTROL ASSEMBLY 2453A05			
1/1	15280-507-41	94756	POWER CONTROL ASSEMBLY	REF	A	PAFDA
1/1	15280-507-61	94756	POWER CONTROL ASSEMBLY	REF	B	PAFDA
1/1	15280-507-71	94756	POWER CONTROL ASSEMBLY	REF	C	PAFDA
1/1	15280-507-81	94756	POWER CONTROL ASSEMBLY	REF	D	PAFDA
1/1	15280-507-91	94756	POWER CONTROL ASSEMBLY	REF	E	PAFDA
1/1	15280-507-101	94756	POWER CONTROL ASSEMBLY	REF	F	PAFDA
2/1	15404-507-41	94756	. CHASSIS ASSEMBLY, INNER (FOR . . . . . DETAILS, SEE FIG. 2)	REF	A	XB
2/1	15404-507-61	94756	. CHASSIS ASSEMBLY, INNER (FOR . . . . . DETAILS, SEE FIG. 2)	REF	B	XB
2/1	15404-507-71	94756	. CHASSIS ASSEMBLY, INNER (FOR . . . . . DETAILS, SEE FIG. 2)	REF	C	XB
2/1	15404-507-81	94756	. CHASSIS ASSEMBLY, INNER (FOR . . . . . DETAILS, SEE FIG. 2)	REF	D	XB
2/1	15404-507-91	94756	. CHASSIS ASSEMBLY, INNER (FOR . . . . . DETAILS, SEE FIG. 2)	REF	E	XB
2/1	15404-507-101	94756	. CHASSIS ASSEMBLY, INNER (FOR . . . . . DETAILS, SEE FIG. 2)	REF	F	XB
3/1	15400-507-11	94756	. CHASSIS AND SLIDE ASSEMBLY, ... OUTER (FOR DETAILS, SEE FIG. 3)	REF	G	XB
3/1	15400-507-31	94756	. CHASSIS AND SLIDE ASSEMBLY, ... OUTER (FOR DETAILS, SEE FIG. 3)	REF	H	XB
- 4	15424-507-41	94756	. WIRE HARNESS ASSEMBLY (FOR . . . . . DETAILS, SEE FIG. 4)	REF	A	XB
- 4	15424-507-61	94756	. WIRE HARNESS ASSEMBLY (FOR . . . . . DETAILS, SEE FIG. 4)	REF	B	XB
- 4	15424-507-71	94756	. WIRE HARNESS ASSEMBLY (FOR . . . . . DETAILS, SEE FIG. 4)	REF	C	XB
- 4	15424-507-81	94756	. WIRE HARNESS ASSEMBLY (FOR . . . . . DETAILS, SEE FIG. 4)	REF	D	XB
- 4	15424-507-91	94756	. WIRE HARNESS ASSEMBLY (FOR . . . . . DETAILS, SEE FIG. 4)	REF	E	XB
- 4	15424-507-101	94756	. WIRE HARNESS ASSEMBLY (FOR . . . . . DETAILS, SEE FIG. 4)	REF	F	XB
5/1	14320-507-1	94756	. SERIAL DIGITAL MULTIPLEX . . . . . ASSEMBLY (SDMA)	1		PAFDA
6/1	14320-507-1	94756	. COVER, CHASSIS OUTER	1		XB

- Not Illustrated

FIGURE 3. Typical main assemblies listing - Continued.

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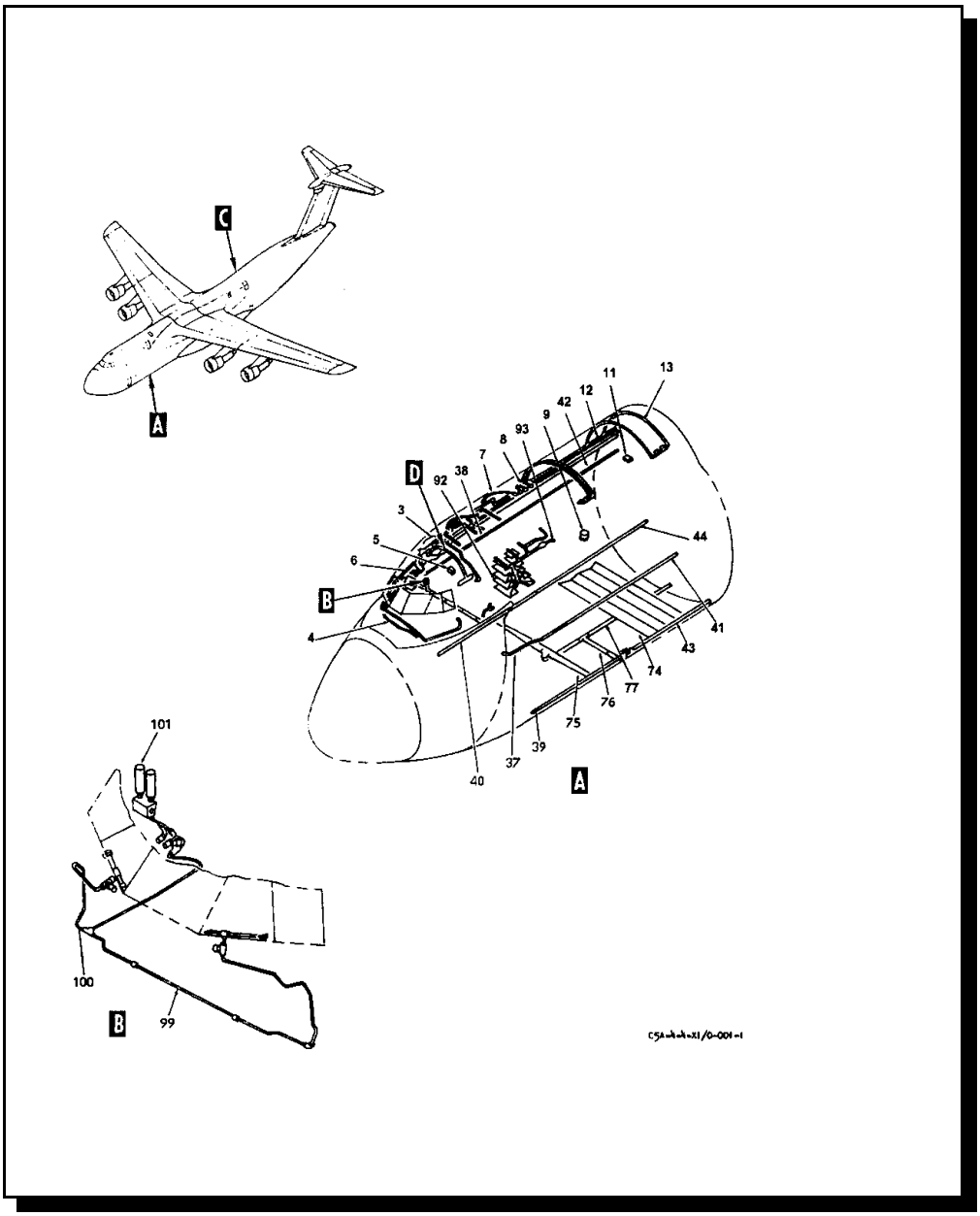


FIGURE 4. Typical functional system listing.

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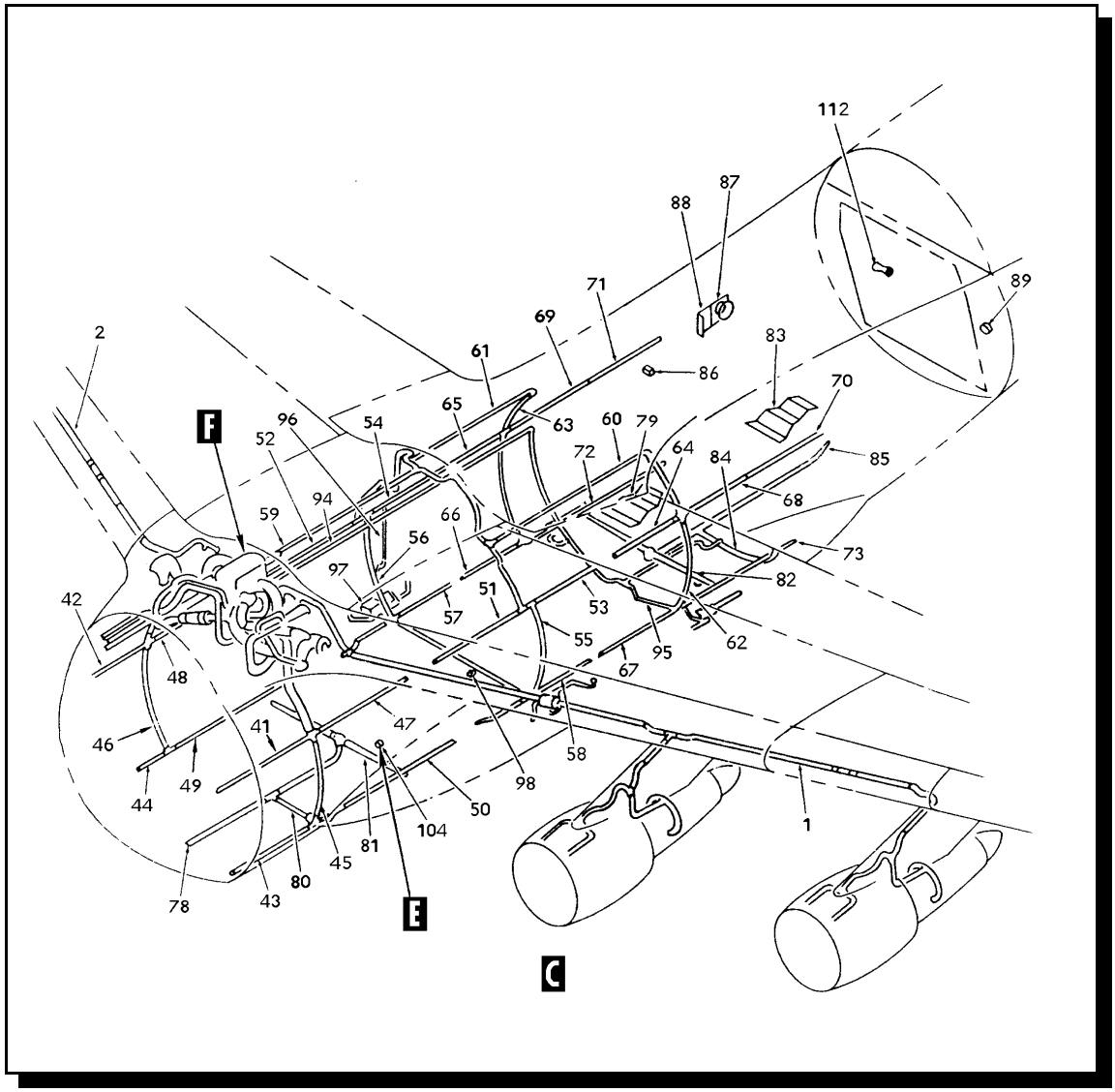


FIGURE 4. Typical functional system listing - Continued.

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Check the source to verify that this is the current version before use.

FIGURE & INDEX / SHEET NO.	PART NUMBER	CAGE	1 2 3 4 5 6 7 DESCRIPTION	UNITS PER ASSY	USEABLE ON CODE	SMR CODE
4- 2 - 29/3	4A10044-101A		. . TRANSMITTER INSTL, PRESSURE, BLEED AIR MANIFOLD, FUSELAGE STATION 1084.000 (FIG. 014)	NP		XC
30/3	4A10044-102A		. . TRANSMITTER INSTL, PRESSURE, BLEED AIR MANIFOLD, FUSELAGE STATION 1084.000 (FIG. 014)	NP		XC
31/3	4A10014-101A		. . VALVE INSTL, FLOW CONTROL, AIR CONDITIONING COMPARTMENT (FIG. 014) (MODIFIED BY 4A19011-101A AND -102A)	NP		XC
32/3	4A10014-102A		. . VALVE INSTL, FLOW CONTROL, AIR CONDITIONING COMPARTMENT (FIG. 014) (MODIFIED BY 4A19011-101A AND -102A)	NP		XC
33/3	4A10010-101A		. . BAFFLE INSTL, AIR CONDITIONING SYSTEM ISOLATOR (FIG. 014)	NP		XC
34/3	4A54000-101A		. . DUCT INSTL, BLEED AIR MANIFOLD, CENTER WING (FIG. 014)	NP		XC
35/3	4A10012-103A		. . TUBE INSTL, OVERBOARD DRAIN, WATER SEPARATOR AND BLEED AIR DUCTS (FIG. 014)	NP		XC
36/3	4A10185-101A		. . BRACKET INSTL, SUPPORT, ENVIRONMENTAL EQUIPMENT, FUSELAGE STATION 1024.000 TO 1084.000 (FIG. 014)	NP		XC
37/1	4A11101-101A		. . DUCT INSTL, AIR DISTRIBUTION, UPPER, FORWARD CARGO COMPARTMENT, FUSELAGE STATION 470.000 TO 636.880 (FIG. 015)	NP		XC
38/1	4A11101-103A		. . DUCT INSTL, AIR DISTRIBUTION, UPPER, FORWARD CARGO COMPARTMENT, FUSELAGE STATION 470.000 TO 636.880 (FIG. 015)	NP		XC
39/1	4A11103-103A		. . DUCT INSTL, AIR DISTRIBUTION, CARGO COMPARTMENT, FUSELAGE STATION 543.000 TO 776.675 (FIG. 015)	NP		XC
40/1	4A11103-105A		. . DUCT INSTL, AIR DISTRIBUTION, CARGO COMPARTMENT, FUSELAGE STATION 543.000 TO 776.675 (FIG. 015)	NP		XC
41/1+	4A11102-101A		. . DUCT INSTL, AIR DISTRIBUTION, UPPER, FORWARD CARGO COMPARTMENT, FUSELAGE STATION 636.880 TO 1056.800 (FIG. 015)	NP		XC
42/1+	4A11102-102A		. . DUCT INSTL, AIR DISTRIBUTION, UPPER, FORWARD CARGO COMPARTMENT, FUSELAGE STATION 636.880 TO 1056.800 (FIG. 015)	NP		XC
43/1+	4A11104-101A		. . DUCT INSTL, AIR DISTRIBUTION, CARGO COMPARTMENT, FUSELAGE STATION 776.675 TO 1056.800 (FIG. 015)	NP		XC
44/1+	4A11104-102A		. . DUCT INSTL, AIR DISTRIBUTION, CARGO COMPARTMENT, FUSELAGE STATION 776.675 TO 1056.800 (FIG. 015)	NP		XC
45/2	4A11100-101A		. . DUCT INSTL, AIR DISTRIBUTION, FORWARD CARGO COMPARTMENT, FUSELAGE STATION 1073.800 (FIG. 015)	NP		XC
46/2	4A11100-102A		. . DUCT INSTL, AIR DISTRIBUTION, FORWARD CARGO COMPARTMENT, FUSELAGE STATION 1073.800 (FIG. 015)	NP		XC
47/2	4A12100-101A		. . DUCT INSTL, UPPER, CENTER CARGO COMPARTMENT, AIR DISTRIBUTION, FUSELAGE STATION 1088.560 TO 1208.500 (FIG.016)	NP		XC

FIGURE 4. Typical functional system listing - Continued.

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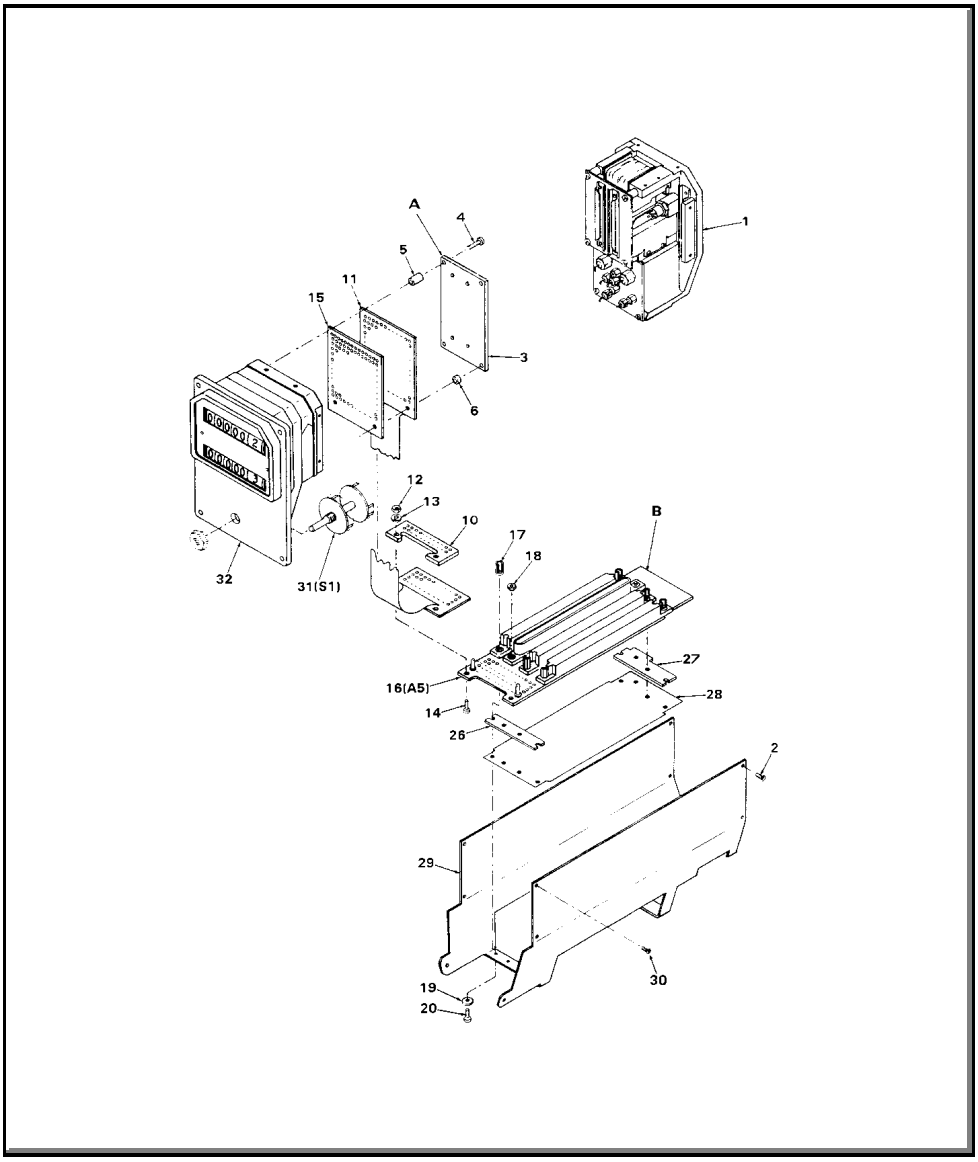


FIGURE 5. Typical exploded view.

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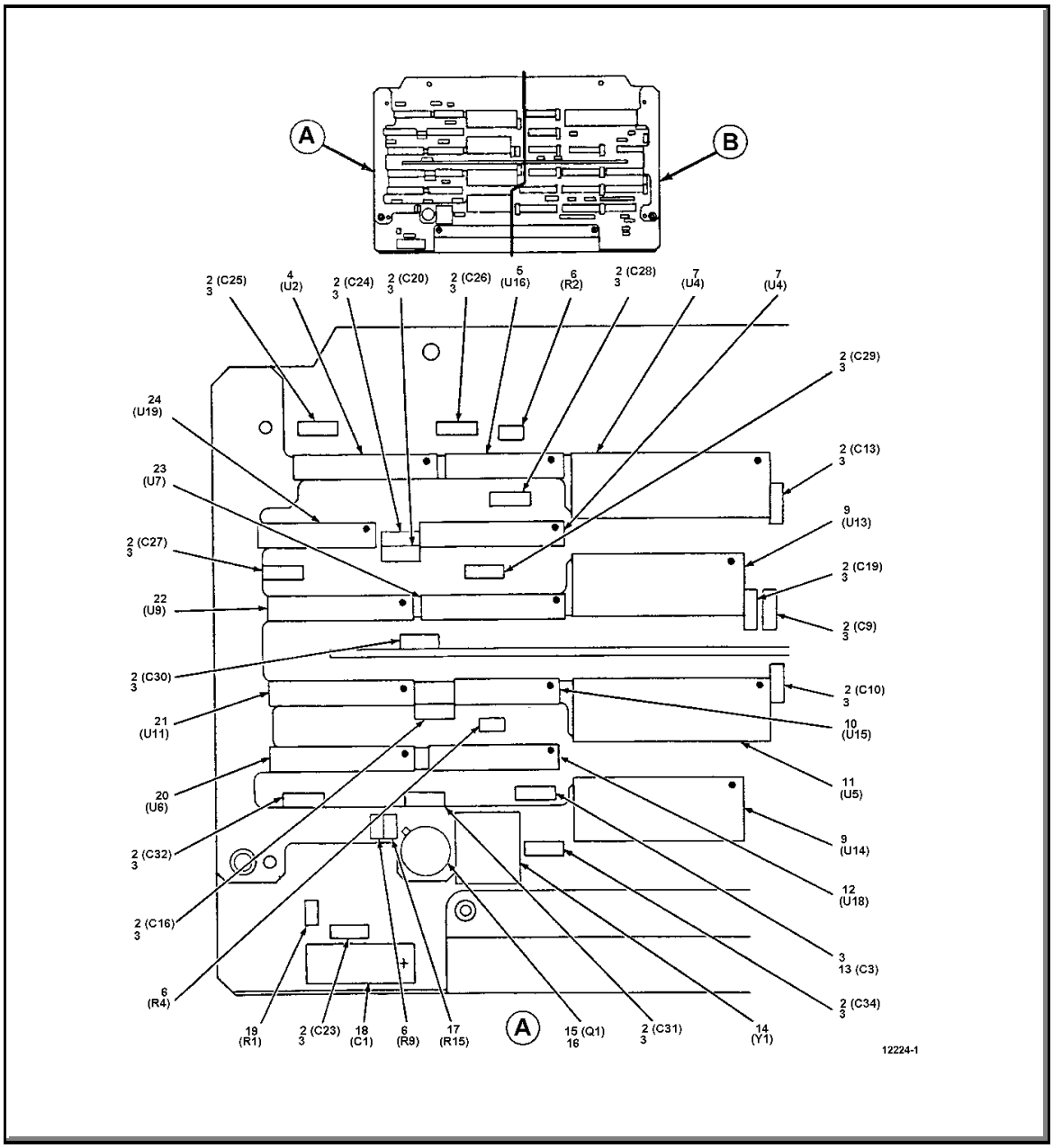


FIGURE 6. Typical circuit card assembly listing.

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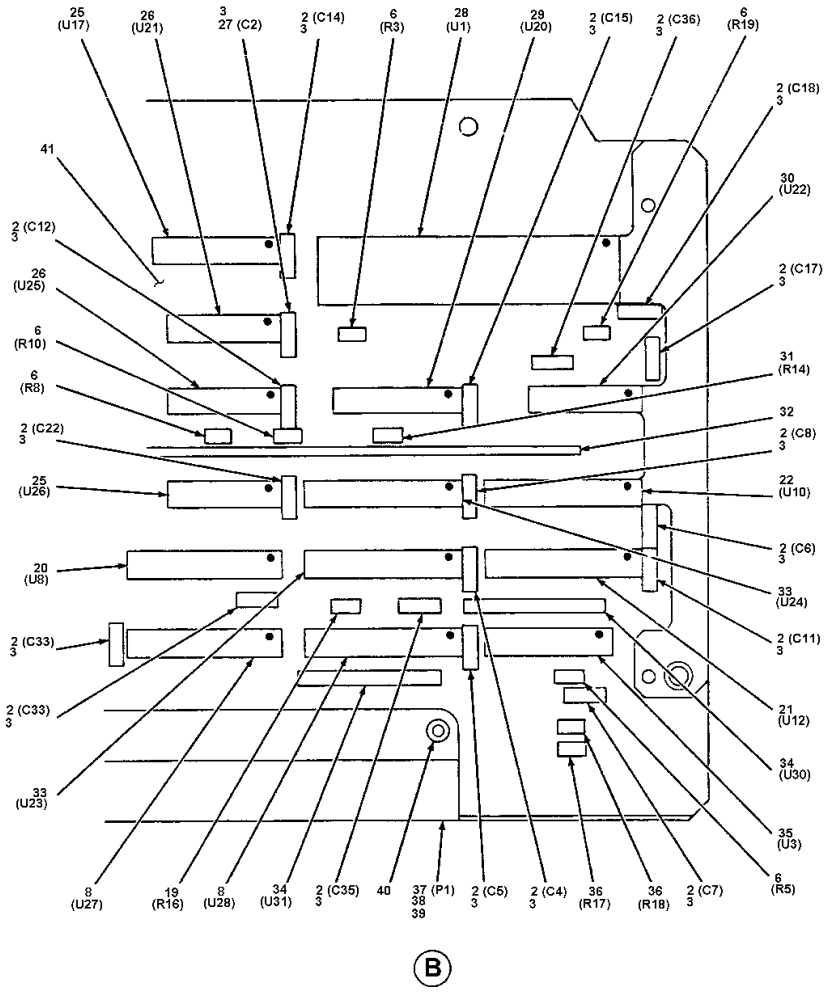


FIGURE 6. Typical circuit card assembly listing - Continued.

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FIGURE & INDEX / SHEET NO.	PART NUMBER		CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USEABLE ON CODE	SMR CODE
2				CIRCUIT CARD ASSEMBLY, BIU NO. 2.			
1	562-0038-003		78711	CIRCUIT CARD ASSEMBLY, BIU NO. 2 . . . . . (FOR NHA SEE FIG. 1)	REF		PAFLD
2/1	M39014/02-1230		81349	. CAPACITOR	33		PADZZ
3/1	302-200		32559	. SPACER, COMPONENT (TELEPHONICS . . . . . SPEC CONTROL S235A202-3)	35		PADZZ
4/1	MD8288/883B		34649	. MICROCIRCUIT, DIGITAL (TELEPHONICS . . . . . SPEC CONTROL M235A716-1)	1		PADZZ
5/1	MM54C163J/883B		27014	. MICROCIRCUIT, DIGITAL (TELEPHONICS	1		PADZZ
	CD40163BMJ/883	=	27014	SPEC CONTROL M235A720-1)			PADZZ
	CD40163BF/3	=	54590				PADZZ
6/1	RLR05C1002GM		81349	. RESISTOR	8		PADZZ
7/1	562-9001-001		78711	. INTEGRATED CIRCUIT, PROGRAMMED (MAKE FROM 14933 PN DESC8200504YX)	1		MDD
8/1	MM54C941J/883B		27014	. MICROCIRCUIT, LINEAR (TELEPHONICS . . . . . SPEC CONTROL M235A702-1)	1		PADZZ
9/1	1DT-6116SA-90DB		61772	. MICROCIRCUIT, DIGITAL (TELEPHONICS . . . . . SPEC CONTROL M235A782-3)	2		PADZZ
10/1	M38510/32702BCX		81349	. MICROCIRCUIT	1		PADZZ
11/1	562-9001-002		78711	. INTEGRATED CIRCUIT, PROGRAMMED (MAKE FROM 14933 . . PN DESC8200504YX)	1		MDD
12/1	MD8284/B		34649	. MICROCIRCUIT, LINEAR (TELEPHONICS . . . . . SPEC CONTROL M235A538-1)	1		PADZZ
13/1	M39014/01-1203		81349	. CAPACITOR	1		PADZZ
14/1	174-21-1		00135	. CRYSTAL, 15 MHZ (TELEPHONICS SPEC	1		PADZZ
	VM6-1500	=	21821	CONTROL C235A374-1)			PADZZ
	80AX1260-2	=	57481				PADZZ
15/1	T235A331-2		78711	. TRANSISTOR, FIELD EFFECT (FET)	1		PADZZ
16/1	M38527/8-40P		81349	. MOUNTING PAD	1		PADZZ
17/1	RLR05C1001GM		81349	. RESISTOR			PADZZ
18/1	M39003/01-2273		81349	. CAPACITOR	1		PADZZ
19/1	RLR05C1001GM		81349	. RESISTOR	2		PADZZ
20/1	560-9003-001		78711	. INTEGRATED CIRCUIT, PROGRAMMED (MAKE . . . . . FROM 14933 PN DESC8103602RX)	1		MDD
21/1	M38510/32502BRX		81349	. MICROCIRCUIT	2		PADZZ
22/1	M38510/32803BRX		81349	. MICROCIRCUIT	2		PADZZ
23/1	MM54C240J/883B		27014	. MICROCIRCUIT, LINEAR (TELEPHONICS . . . . . SPEC CONTROL M235A710-1)	2		PADZZ
24/1	M38510/30702BEX		81349	. MICROCIRCUIT	1		PADZZ
25/2	M235A688-1		78711	. MICROCIRCUIT, DIGITAL	1		PADZZ
26/2	M38510/30501BCX		81349	. MICROCIRCUIT	3		PADZZ
27/2	M39014/02-1218		81349	. CAPACITOR	1		PADZZ
28/2	M235A715-5		78711	. MICROCIRCUIT, DIGITAL	1		PADZZ
	M235A715-1	=	78711				PADZZ
29/2	M38510/30701BEX		81349	. MICROCIRCUIT	1		PADZZ
30/2	M38510/31004BCX		81349	. MICROCIRCUIT	1		PADZZ
31/2	RLR05C5101GM		81349	. RESISTOR	1		PADZZ

FIGURE 6. Typical circuit card assembly listing - Continued.

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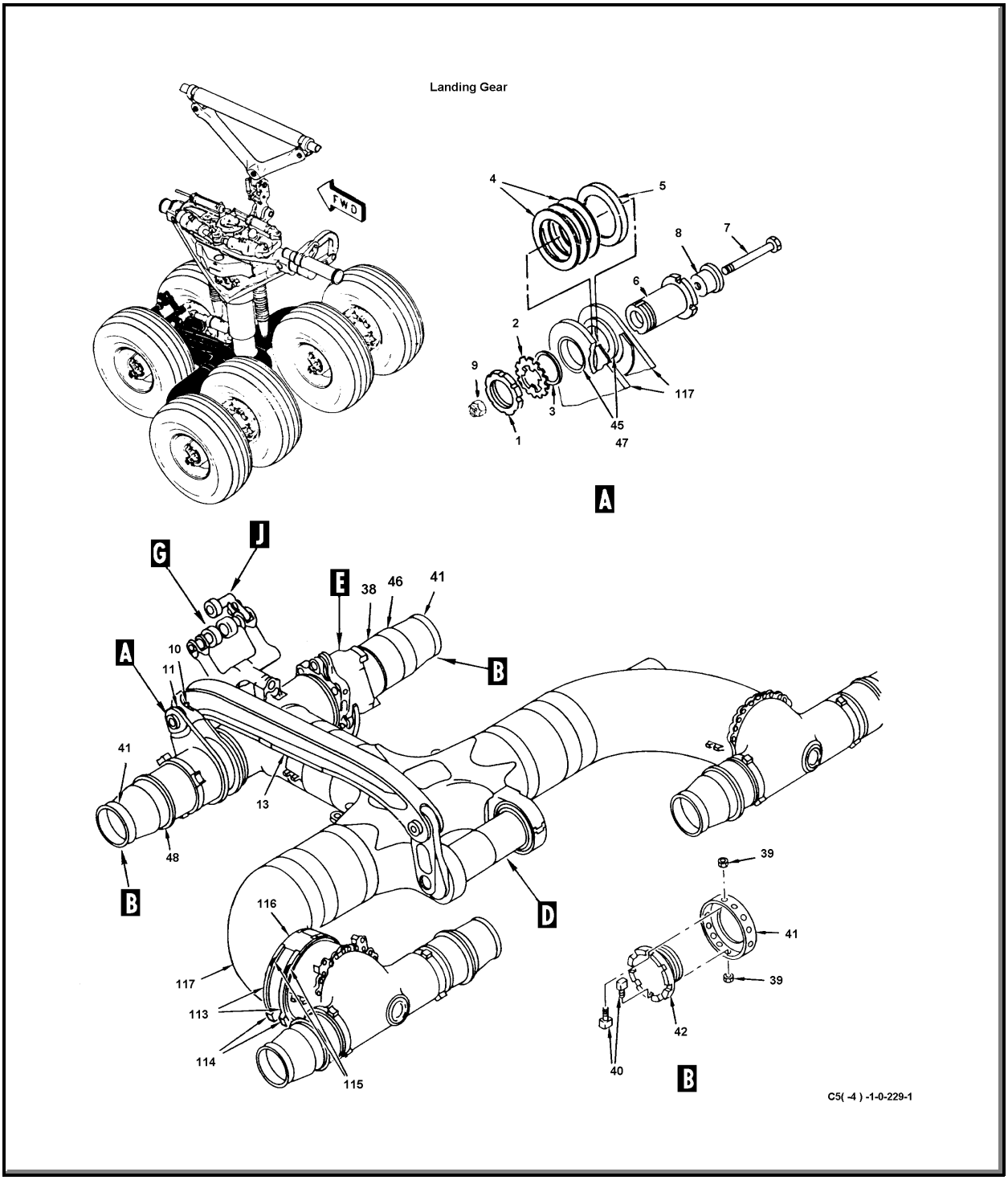


FIGURE 7. Typical MPL listing.

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FIGURE & INDEX / SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USEABLE ON CODE	SMR CODE
1- 242 -			MAIN LANDING GEAR BOGIE			
	4G11014-101B#		. BOGIE, BEAM, MAIN LANDING GEAR (LH) (NHA FIG. 235 AND 236)	REF		PAODD
	4G11014-102B#		. BOGIE, BEAM, MAIN LANDING GEAR (RH) (NHA FIG. 235 AND 236)	REF		AOO
1/1+	4G19068-101A		. . NUT, SPECIAL, BRAKE TORQUE LINK PIN, MAIN LANDING GEAR (USED WITH 4G19067-101A, 4G19069-101A AND 4G19070-101A AS A GROUP)	2		PAOZZ
2/1+	4G10969-101A		. . WASHER, LOCKNUT, BRAKE TORQUE COMPENSATOR PIN, MAIN LANDING GEAR (USED WITH 4G10967-101A, 4G109068-101A AND 4G19070-101A AS A GROUP)	2		PAOZZ
3/1+	4G13554-101A		. . WASHER, THRUST, AXLE BEAM ASSY, MAIN LANDING GEAR	2		PAOZZ
4/1+	4G14504-101A		. . WASHER, BRAKE, TORQUE COMPENSATOR LINK, MAIN LANDING GEAR	AR		PAOZZ
5/1	4G14503-101A		. . WASHER ASSY, PAD, BRAKE, TORQUE COMPENSATOR LINK, MAIN LANDING GEAR (MAY BE USED WHEN 4G14503-107A NOT AVAILABLE)	1	B	PAOZZ
5/1	4G14503-107A		. . WASHER ASSY OF, PAD, BRAKE, TORQUE COMPENSATOR LINK, MAIN LANDING GEAR (REPLACEMENT FOR 4G14503-101A)	1	C	PAOZZ
6/1	4G19067-101A		. . PIN, EXTERNAL THREAD, BRAKE TORQUE COMPENSATOR LINK, MAIN LANDING GEAR (USED WITH 4G19068-101A, 4G10969-101A AND 4G19070-101A AS A GROUP)	2		PAOZZ
7/1	NAS1004-52		. . BOLT	1		PAOZZ
8/1	4G19070-101A		. . ADAPTER, BRACKET RETAINER, AXLE BEAM ASSY MAIN LANDING GEAR (USED WITH 4G19067-101A, 4G19068-101A AND 4G19069-101A AS A GROUP	1		PAOZZ
9/1	52-1650-048		. . NUT, SELF-LOCKING, FLUSH, 100 DEGREE, SLOTTED, 250 DEGREE FAHRENHEIT (72962) (LOCKHEED SPEC DWG STSCJ011-08)	1		PAOZZ
	4G11017-1018		. . LINK ASSY, BRAKE, TORQUE COMPENSATING, MAIN LANDING GEAR	1		PAOD
10/1	MS15001-1		. . . FITTING	1		PAOZZ
11/1	KWB28-7		. . . BEARING (97613)			
12/1	4G14128-107A		. . . BUSHING, FLANGED, MAIN LANDING GEAR BOGIE (PREFERRED SPARE IN LIEU OF 4G14128-105A)	1		
13/1+	4G11017-103B		. . . LINK, BRAKE, TORQUE COMPENSATING, MAIN LANDING GEAR	1		XA
14/4	4G13850-103A		. . BOLT, SPECIAL, AXLE BEAM, MAIN LANDING GEAR	1		PAOZZ
15/4	4G13851-101A		. . WASHER, SPECIAL, AXLE BEAM, MAIN LANDING GEAR	1		PAOZZ
16/4	MS21046C7		. . NUT	1		PAOZZ
17/4	4G13852-101B		. . BUSHING, CLAMP UP, AXLE BEAM MAIN LANDING GEAR	1		PAOZZ
18/4	4G12414-101A		. . ROLLER ASSY, LOCK, MAIN LANDING GEAR	1		PAOZZ
	4G12415-101C		. . FITTING ASSY, LOCK ROLLER AXLE BEAM ASSEMBLY, MAIN LANDING GEAR	1		PAOLD
19/4	4G13849-105A		. . . BUSHING, AXLE BEAM ASSY, MAIN LANDING GEAR (SPARES ONLY) (PREFERRED SPARE IN LIEU OF 4G13849-101A AND -103A)	1		PAOZZ
20/4	4G13848-105A		. . . BUSHING, AXLE BEAM ASSY, MAIN LANDING GEAR (SPARES ONLY) (PREFERRED SPARE IN LIEU OF 4G13848-101A AND -103A)	1		PAOZZ

FIGURE 7. Typical MPL listing - Continued.

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FIGURE & INDEX / SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USEABLE ON CODE	SMR CODE
1-242 - 111/2	KWB28-7		. . . BEARING (97613)	1		PAOZZ
112/2	4G11018-103B		. . . GUDGEON, AXLE BEAM, MAIN LANDING GEAR	1		PAOZZ
113/2	4G11014-109A		. . STRAP, AXLE, FORWARD, AXLE BEAM ASSY, MAIN LANDING GEAR	2		PAOZZ
114/1	1522		. . SEAL (00266)	2		PAOZZ
115/1	4G13085-127A		. . PAD, CLAMP, HYDRAULIC LINE ROUTE, MAIN LANDING GEAR (PFD SPARE IN LIEU OF 4G13085-111A)	2		MOO
116/1	4G14380-101A*		. . PLATE, CONFIGURATION, BOGIE BEAM ASSY MAIN LANDING GEAR	1		MDO
117/1+	4G11014-103C		. . BOGIE ASSY, BEAM, MAIN LANDING GEAR (FIG. 243)	1		PAODD
	4G11105-101A		. LANDING GEAR BUILT UP UNIT, MAIN, LH AFT (SEE TO 4S1-93-12) (NHA FIG. 236) (PFD REPLACEMENT FOR 4G11005-113A)	1	D	AOO
	4G11105-103A		. LANDING GEAR BUILT UP UNIT, MAIN, RH AFT (SEE TO 4S1-93-12) (NHA FIG. 236) (PFD REPLACEMENT FOR 4G11005-115A)	1	D	AOO
118/4	4G13924-101A		. . PIN, TRUNNION, MASTER CYLINDER, MAIN	2	D	PAOZZ
			LANDING GEAR A AF66-8304 THRU AF70-467 B AF66-8304 THRU AF85-0008 C AF85-0009 AND UP D AF66-8304 THRU AF70-467 AFTER TO 1C-5A-1978 AF83-1285 AND UP			

FIGURE 7. Typical MPL listing - Continued.

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APPENDIX A

ILLUSTRATED PARTS BREAKDOWN
DOCUMENT TYPE DEFINITION (DTD) SUBSET

A.1 SCOPE.

A.1.1 Scope. The markup tags described herein are based on rules outlined in the Information Processing, Text and Office Systems, Standard Generalized Markup Language (SGML) Standard, International Organization for Standardization (ISO) 8879, as incorporated in Federal Information Processing Standards (FIPS) PUB 152, and MIL-PRF-28001. The Document Type Definition (DTD) subset within this appendix provides the structure and content of documents prepared in accordance with this specification. Digital copies of the DTD (see A.4.1), tag description table (see A.4.2), attribute description table (see A.4.3), and the Formatting Output Specification Instance (FOSI) (see A.5) are available (see A.6). The Tag Description table provides a detailed discussion of each markup tag. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS.

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (see 6.2)

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-28001	-	Markup Requirements and Generic Style Specification for Electronic Printed Output and Exchange of Text

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-2171.)

STANDARDS

FEDERAL INFORMATION PROCESSING STANDARDS

FIPS PUB 152	-	Standard Generalized Markup Language (SGML)

(Copies of FIPS are available to Department of Defense activities from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Others must request copies of FIPS from the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161-2171.)

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APPENDIX A

A.3 DOCUMENT TYPE DEFINITION SUBSET.

A.3.1 SGML document type definition subset. Data to be delivered digitally in accordance with this specification shall be SGML tagged using the DTD found in MIL-STD-38784 as modified by the DTD subset in this section. The procedure for accomplishing this is found in MIL-PRF-28001 and FIPS PUB 152 (ISO 8879).

A.3.2 Subdivided IPB manuals. If an IPB manual is to be subdivided, it will be necessary to revise the DTD subset. By changing the value of the "subdivide" entity to "include", an IPB can be subdivided. If the DTD subset is modified for a subdivided manual, the data and the DTD shall be delivered in accordance with MIL-STD-1840.

A.3.3 Template document type for Illustrated Parts Breakdown. The DTD subset for the Operation and Maintenance Work Packages DTD is as follows:

```
<!-- ************************ START OF FILE ************************** -->

<!-- DOCUMENTATION -->

<!-- 12/5/96  mlv

This is for the Combined Manual (38784 & 38807) it needs to be set to IGNORE
if not using it for a Combined Manual, and set to INCLUDE for an IPB manual or
WP
-->

<!--
<!ENTITY % ipbchapter "INCLUDE" >
<![ %ipbchapter; [

<!ENTITY % m38784comb PUBLIC "-//USA-DOD//DTD MIL-STD-38784 AMEND1 DRAFT//EN">

%m38784comb;

]]>

-->

<!-- SUPPLEMENT NOTICE: This file is made available to provide the user with
a digital representation of the DTD found in Appendix A of MIL-PRF-38807C
Amendment One.

This file is incomplete without MIL-PRF-38807C -->

<!-- NOTE: The start and end of this file are marked with a row of asterisks.
If these rows are not present the file may not be complete! -->

<!-- MIL-PRF-38807C IPB AMEND1 DRAFT DTD -->

<!-- The following set of declarations may be referred to by using a
public entity as follows:

<!ENTITY % m38807cPRF PUBLIC
    "-//USA-DOD//DTD MIL-PRF-38807C IPB AMEND1 DRAFT//EN" >
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APPENDIX A

```
%m38807;
-->

<!-- NOTE: In order to parse the following DTD subset alone, append
the following statement to the beginning of the file:

     <!DOCTYPE docipb [

and the associated "]>" to the end of the file. -->

<!-- ENTITY DECLARATIONS -->


<!ENTITY % frnt "(idinfo, lep, verstat?, contents)" >

<!ENTITY % mpl "((figindex | assy), refdes*, sssn*, (partno, cage, desc,
units, useoncode?, smrcodes)+)+" >

<!-- For pl0 (%mpl;) with no 'figindex' number, the author should enter an
'<assy>' tag to the tagged instance.  This will indicate that the figindex
column should be empty. The figure number should appear in this column
automatically. -->

<!ENTITY % subdivide "IGNORE" >

<![ %subdivide; [

<!ENTITY % bodyele "(ipbforeword?, pl?, numindx?, rfdsindx?)" >

]]>

<!ENTITY % bodyele "(ipbforeword, pl, numindx?, rfdsindx?)" >

<!-- Since numindx and rfdsindx are empty, it would be unwise to use this
type of strategy to get multiple manuals.  Using a document with,
a body of <numindx>' would yield a chapter title a header only.  Since
all of the relevant information is contained in the '<pl>' a different
approach is needed.  -->

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attachpart %yesorno; '0'
marking %yesorno; '0'
kit %yesorno; '0'
quickchgexpl %yesorno; '0' " >

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<!ENTITY % parazero "(%titles;, warning*, caution?, note?, (para,
note?)?, (step1, step1+)?)" >

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                      tmidno, title)+)" >

<!ENTITY % symsect "(%parazero;, subpara1*, (title?, (term, def)+)?)" >
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APPENDIX A

```
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manstat (comb | single) 'single'
verstatpg (all | unverified) 'all'" >

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quickchgexpl?, symsect, abbrsect?, shtnoexpl?, useonexpl, smrexpl, ipbhci?,
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para0*)" >

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sssn CDATA #IMPLIED
indexid  CDATA  #IMPLIED
%chgatt;
%db; " >

<!ENTITY % para0att
"shortentry %yesorno; '0'
emergency %yesorno; '0'
id ID #IMPLIED
```


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```
hcp %yesorno; '0'
hci %yesorno; '0'
esds %yesorno; '0'
nsp %yesorno; '0'
applicref IDREFS #IMPLIED
%verified;
%bodyatt;" >

<!-- This is for the Combined Manual (38784 & 38807) it needs to be set to
IGNORE if using it for a Combined Manual, and set to INCLUDE for an IPB manual
or WP -->

<!ENTITY % ipbcall "INCLUDE" >
<![ %ipbcall; [

<!ENTITY % m38784 PUBLIC "-//USA-DOD//DTD MIL-STD-38784 AMEND1 DRAFT//EN">

%m38784;

]]>

<!-- ELEMENT and ATTRIBUTE LIST DECLARATIONS -->

<!ELEMENT assy         - o  EMPTY >

<!ELEMENT cage         - o  (%text;) >

<!ELEMENT desc         - o  (partname, remarks?) >

<!ELEMENT docipb       - -  (front, body)   +(line | modreq | subjinfo | pgbrk
                            | brk) >
<!ATTLIST docipb     service %service; 'AF'
                       %docatt; >

<!ELEMENT figindex     - o  (%text;) >


<!ELEMENT finding      - o  (%parazero;, subpara1*) >
<!ATTLIST finding      %para0att; >

<!ELEMENT group10      - o  EMPTY >

<!ELEMENT hci          - o  EMPTY >

<!ELEMENT ipbesds      - o  (%parazero;, subpara1*) >
<!ATTLIST ipbesds      %para0att; >

<!ELEMENT ipbhci       - o  (%parazero;, subpara1*) >
<!ATTLIST ipbhci       %para0att; >

<!ELEMENT ipbforeword   - -  (%ipbforwd;)   +(figure | table) >
<!ATTLIST ipbforeword   %chapatt; >

<!ELEMENT mfrlist       - o  (%parazero;, subpara1*) >
<!ATTLIST mfrlist         %para0att; >

<!ELEMENT modelcover    - o  (%parazero;, subpara1*) >
<!ATTLIST modelcover     %para0att; >
```


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APPENDIX A

```
<!ELEMENT numindx      - o  (numgrp)  +(group10) >

<!ATTLIST numindx    cols (2 | 3) "3"
                     autobuild NUMBER #CONREF
                     emergency %yesorno; '0'
           id ID #IMPLIED
           label CDATA #IMPLIED
           sssn CDATA #IMPLIED
           indexid  CDATA  #IMPLIED
           %db;
           %secur; >

<!ELEMENT numgrp       - o  ((partno, figindex, units?)+) >

<!ELEMENT partname     - o  (%text;) >
<!ATTLIST partname     %secur; >

<!ELEMENT partstd      - o  (%parazero;, subpara1*) >
<!ATTLIST partstd      %para0att; >

<!ELEMENT pl           - -  (plgrp+) +(plbrk)>
<!ATTLIST pl           %chapatt; >

<!ELEMENT plbrk        - o  EMPTY >

<!ELEMENT plgrp        - -  (figure, pl0tbl) >

<!ELEMENT pl0tbl       - o  ((%titles;)?, pl0) >

<!ELEMENT pl0          - -  (%mpl;, pl1*, useoncodelist?) >
<!ATTLIST pl0          %verified;
                       %bodyatt;
                       %mplatts; >

<!ELEMENT pl1          - o  (%mpl;, pl2*) >
<!ATTLIST pl1          %mplatts;
                       %bodyatt; >


<!ELEMENT pl2          - o  (%mpl;, pl3*) >
<!ATTLIST pl2          %mplatts;
                       %bodyatt; >

<!ELEMENT pl3          - o  (%mpl;, pl4*) >
<!ATTLIST pl3          %mplatts;
                       %bodyatt; >


<!ELEMENT pl4          - o  (%mpl;, pl5*) >
<!ATTLIST pl4          %mplatts;
                       %bodyatt; >

<!ELEMENT pl5          - o  (%mpl;, pl6*) >
<!ATTLIST pl5          %mplatts;
                       %bodyatt; >

<!ELEMENT pl6          - o  (%mpl;, pl7*) >
```


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APPENDIX A

```
<!ATTLIST pl6          %mplatts;
                       %bodyatt; >

<!ELEMENT pl7          - o  (%mpl;) >
<!ATTLIST pl7          %mplatts;
                       %bodyatt; >

<!ELEMENT quickchgexpl  - o  (%parazero;, subpara1*) >
<!ATTLIST quickchgexpl  %para0att; >

<!ELEMENT refdes       - o  (%text;) >
<!ATTLIST refdes       %secur; >

<!ELEMENT rfdsindx     - o  (rfdsgrp)  +(group10) >
<!ATTLIST rfdsindx     autobuild NUMBER #CONREF
                       type (refdes | sssn | both) "refdes"
                       cols (2 | 3) "3"
                       %chapatt; >

<!ELEMENT rfdsgrp      -  o  ((refdes?, sssn?, figindex)+) >

<!ELEMENT serialization    - o  (%parazero;, subpara1*, sertable) >
<!ATTLIST serialization    %para0att; >

<!ELEMENT sertable     - -  (title?, (nomen, modelno, serno+)+) >
<!ATTLIST sertable     col1width CDATA "1.25in"
                       col2width CDATA "1in"
-- column #3 will be the remainder of the column not used by cols 1 & 2 --
                       %bodyatt; >

<!ELEMENT shtnoexpl    - o  (%parazero;, subpara1*) >
<!ATTLIST shtnoexpl    %para0att; >

<!ELEMENT simassem     - o  (%parazero;, subpara1*) >
<!ATTLIST simassem     %para0att; >

<!ELEMENT smrcode      - o  (%text;) >
<!ATTLIST smrcode      service CDATA #IMPLIED >

<!ELEMENT smrcodes     - o  (smrcode)+ >

<!ELEMENT smrexpl      - o  (%parazero;, subpara1*) >
<!ATTLIST smrexpl      %para0att; >

<!ELEMENT sssn             - o  (%text;) >
<!ATTLIST sssn             %secur; >

<!ELEMENT units            - o  (%text;) >
<!ATTLIST units            %secur; >

<!ELEMENT useoncode        - o  (%text;) >
<!ATTLIST useoncode        %secur; >

<!ELEMENT useoncodelist    - -  (term, def)+   +(note) >
<!ATTLIST useoncodelist    %bodyatt; >

<!ELEMENT useonexpl        - o  (%parazero;, subpara1*) >
<!ATTLIST useonexpl        %para0att; >
```


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```
<!-- ***************************** END OF FILE ***************************** -->
```

A.4 DETAILED DESCRIPTION

A.4.1 Document type definition. The DTD subset within this appendix provides the structure and content of documents prepared in accordance with this specification. The DTD is available in a digital format. See A.6, for information on obtaining the file.

A.4.2 Tag description table. The Tag Description Table provides detailed descriptions of the tags above. It provides the element tagging structure, full element name, tag minimization requirements, element structure, referencing elements, source paragraph, and attribute descriptions unique to the element. See A.6, for information on obtaining this table.

A.4.3 Attribute description table. This Attribute Description Table provides detailed descriptions of the attributes above. See A.6 for information on obtaining this table.

A.5 FORMATTING OUTPUT SPECIFICATION INSTANCE (FOSI).

A.5.1 FOSI description. The FOSI provides instructions to the output system to produce a formatted manual from the tagged instance. The FOSI is provided as Government Furnished Information (GFI). See A.6, for information on obtaining the FOSI.

A.6 OBTAINING FILES.

A.6.1 Obtaining files. The DTD, attribute, and tag description tables are available as ASCII files (see A.6.1.1 and A.6.1.2). In the event of a conflict between the text of this document and any downloaded files, the text of this document takes precedence. These files are for convenience and informational purposes only.

A.6.1.1 File Transfer Protocol (FTP). The procedures for obtaining files via FTP are as follows:

a. Connect to "WPCDSO1.wpafb.af.mil" using the FTP software available at your site. For example, if your FTP software is invoked using the "ftp" command, type "ftp WPCDSO1.wpafb.af.mil". Do not attempt to log-in to this site using a "telnet" connection. If this connection fails, connect using "129.52.152.8".

b. Log-in (login, name, remote user name, etc.) as "ftp" and press "enter".

c. For password, type electronic mail (e-mail) name followed by "@" (at) and press "enter".

d. Type "cd pub/tmss-web" (or the command your system requires to change to "pub/tmss-web" directory) and press "enter". At this point, a short new users message will normally appear. If the new users message does not appear, it should be downloaded and read. Download file by typing "get.message" (or the command your system requires to download a file) and press "enter".

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e. Type "get filelist.txt" (or the command your system requires to download a file) and press "enter". This file contains a list of all files available. This file is updated as new items are added, therefore it should be downloaded and read before downloading any other file.

f. If the needed file ends with ".zip", see g. below, otherwise type "asc" (or the command your system requires for an ASCII transfer) and press "enter". Type "get XXXXXX.XXX" (where XXXXXX.XXX is the name of the file to be downloaded) and press "enter" to download needed file. Repeat for each file to be downloaded.

g. If the needed file ends with ".zip", type "bin" (or the command your system requires for a binary transfer) and press "enter". Type "get XXXXXX.XXX" (where XXXXXX.XXX is the name of the file to be downloaded) and press "enter" to download needed file. Repeat for each file to be downloaded. Zipped files were compressed using PKZIP Version 2.04.

h. File "nc.txt" contains information on the naming conventions used on all files in this directory. Type "get nc.txt" to download this file.

A.6.1.2 World Wide Web (WWW). Obtain files via the Air Force Product Data Systems Modernization (PDSM) Program Office home page at "http://www.pdsm.wpafb.af.mil/." Select TMSS from the graphical menu and follow the directions presented.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waiver any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-38807C (USAF)	96/11/29

3. DOCUMENT TITLE
Technical Manuals - Illustrated Parts Breakdown

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
Steven C. Holloway	(1) Commercial (937) 257-3085	(2) AUTOVON 787-3085
c. ADDRESS *(Include Zip Code)* Det 2, HQ ESC/AV-2C 4027 Col Glenn Hwy, Suite 300 Dayton, OH 45431-1672	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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